Low Computational Complexity RLS-Based Decision-Feedback Equalization in Underwater Acoustic Communications

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***Abstract*—The adaptive recursive least squares (RLS) algorithm plays a crucial role in underwater acoustic (UWA) communications because of its robustness and fast convergence. However, the high computational complexity of the RLS algorithm has limited its application in UWA channels with long delay spreads. Two strategies can be developed to reduce the complexity of the RLS algorithm. The first one is to directly reduce the complexity considering the algorithm itself. Here, we exploit the dichotomous coordinate descent (DCD) algorithm to achieve low complexity. Since the received signals from multiple hydrophones are compensated with fluctuating phase increments at each instant, the shift structure of the input sequence, which enables a significant reduction in the complexity of updating the correlation matrix of the input vector in the DCD algorithm, is no longer applicable in UWA channels. To make the DCD algorithm compatible with the nonshifted structure of input sequence, a partial updating approach is employed for the correlation matrix of the input sequence over time in the RLS algorithm. This approach skips cases with small phase variations and only updates the cross-correlation submatrix. In this way, constant and full-scale phase compensation is avoided. The other strategy is to shorten the channel length. Here, we exploit the iterative frequency-domain equalization (FDE) to suppress the intersymbol interference from multipaths. The received signal is first partitioned into overlapping subblocks for iterative FDE. A weighting processing with a subblock forgetting factor is designed to make the mean squared error continuous across subblocks and iterations in the RLS algorithm. Both the strategies were adopted in the decision-feedback equalization (DFE) and examined by simulations and experiments. Results demonstrate that the proposed algorithms approximate or outperform the traditional RLS-based DFE with much lower computational overheads in channels with small delay spreads and fluctuation rates. For the improved DCD-RLS-based DFE, a threshold provides a tradeoff between the performance and complexity. For the FDE-RLS-based DFE, the mean squared error and computational complexity induced by long equalizer taps can be kept at low levels due to channel shortening. The latter algorithm remains effective even as channel delays and fluctuation rates increase.**

***Index Terms*—Channel shortening, decision-feedback equalization (DFE), dichotomous coordinate descent (DCD), recursive least squares (RLS), underwater acoustic (UWA) communications.**

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I. Introduction

THE recursive least squares (RLS) algorithm plays a crucial role because of its fast convergence in adaptive updating of filters for underwater acoustic (UWA) communications. The high computational complexity, which is $\mathcal{O}[N_0^2]$ per sample (N_0 is the filter length that depends on the channel length), has limited the further application of the RLS algorithm in UWA channels with long delay spreads. To reduce the complexity of the RLS algorithm, two strategies can be developed. One is to improve the algorithm itself and reduce the complexity to $\mathcal{O}[N_0]$; the other is to shorten the channel length to $N_0' \ll N_0$ and, thus, achieve a much lower complexity of $\mathcal{O}[{N_0'}^2]$. Both the strategies are investigated in this article, and effective algorithms are presented for UWA communications.

In the first strategy, several improved filters have been designed to reduce the complexity of the RLS algorithm. Some filters achieve the complexity of $\mathcal{O}[N_0]$ but at the price of performance degradation. The improved adaptive filters can be classified into two categories: 1) the fix-order filters and 2) the recursive-order filters. The shifted structure of the data sequence is exploited in both the types of filters to reduce the complexity. Typical fix-order filters are the fast transversal filter (FTF), the fast a posteriori error sequential technique, and the fast Kalman filter [1]. However, these filters suffer from numerical instabilities as the algebraic equalities that are used to derive the algorithms tend to break down in finite precision. To address the problem, the stabilized FTF method has been proposed [2], [3], which introduces redundancy into the computation of certain quantities, such as the backward prediction error. This redundancy is utilized to estimate the numerical errors accumulated in these quantities, and these errors are, thus, employed in a feedback mechanism to counteract destabilizing effects. Despite the qualification of the stabilized FTF, a careful initialization is required for maintaining numerical stability [4], and the stability can only be guaranteed when the forgetting factor is very close to one [5]. Another method is to insert leakage into some of the FTF equations, but the introduced bias of weight estimates may lead to divergence even at the initial stage of the adaptation [1]. In the recursive-order filters, a notable feature is the lattice algorithm, which is based on the forward and backward linear predictors. However, the lattice algorithm does not provide the filter weights required in many applications, and the complexity is still high [6]. An example is the Kalman gain estimator, which requires $13N_0 \log_2 N_0$ multiplications and $3N_0 \log_2 N_0$ divisions per sample [7]. In 2008, a low-complexity RLS algorithm based on the dichotomous coordinate descent (DCD) iterations was proposed. The DCD-RLS algorithm performs close to the classical RLS algorithm with complexity as low as $3N_0$ multiplications per sample. The sparse counterparts integrating l_0, lasso, ridge regression, and elastic-net penalties have also been investigated for performance improvement [8]. In spite of the excellent performance, its complexity remains high when the DCD-RLS algorithm is applied to the equalization in UWA communications as the shifted structure no longer holds. The received signals from multiple hydrophones should be compensated with fluctuating phase increments at each instant.

In the second strategy, the dominant linear filters for channel shortening are based on time reversal [9] and minimum mean squared error (MMSE) [10]. The time-reversal filter exhibits temporal and spatial focusing characteristics, effectively compressing the intersymbol interference (ISI) [11], [12], [13]. However, the low sidelobe of the Q function can hardly be achieved with limited hydrophones. On the other hand, the MMSE filter is considered the theoretically optimal shortening filter, resulting in a shorter channel compared to the time-reversal filter [14], [15]. As the noise level increases, the MMSE filter converges to the time-reversal filter, leading to a hybrid approach that combines both the algorithms [16]. However, these methods may encounter a performance limitation, motivating the exploration of nonlinear filters for performance enhancement. Despite this, both the methods can be implemented in the frequency domain in a block-by-block manner and merge the multichannel signals from multiple hydrophones for single-channel equalization at a low complexity cost, inspiring us to exploit these features.

In this article, we investigate the low-complexity RLS algorithms in UWA communications. In the strategy of enhancing the RLS algorithm, an improved DCD-RLS (*i*DCD-RLS) algorithm is proposed considering the fluctuating carrier phase characteristic, which makes the fast updating of the correlation matrix in the tradition DCD algorithm inapplicable in UWA channels. In the *i*DCD-RLS algorithm, we employ a partial updating approach for the correlation matrix of the input sequence over time within the RLS framework, selectively updating the cross-correlation submatrix and omitting cases with minor phase variations. As a result, constant and full-scale phase compensation is avoided. It should be noted that our algorithm differs from the one in [17], which utilizes a partial-update method based on DCD-RLS for channel estimation before deriving the decision-feedback equalization (DFE) tap coefficients, categorizing it as a channel-estimate-based DFE. In contrast, our algorithm falls under directly adaptive DFE, directly tracking the DFE tap coefficients using a partial-update method based on the RLS algorithm. This approach reduces computational complexity and minimizes errors in DFE tap coefficient calculations caused by potential channel estimation errors. In the strategy of channel shortening, frequency-domain equalization (FDE) is exploited to suppress the ISI. We integrate the benefits of blockwise FDE and symbolwise RLS-DFE, resulting in a low-complexity FDE-RLS-based DFE method with rapid convergence and reduced mean squared error (MSE). In this method, the received signal is partitioned into overlapping subblocks for iterative FDE. A weighting processing with a subblock forgetting factor is designed to ensure continuous MSE across subblocks and iterations in the RLS algorithm. The performance and complexity of the proposed *i*DCD-RLS- and FDE-RLS-based DFEs are examined by numerical simulations and experimental data in different ocean environments.

The contributions of this article are summarized as follows. First, we develop an improved low-complexity DCD-RLS algorithm for the multichannel adaptive equalization of fast time-varying channels in UWA communications. The partial updating based on the accumulated phase rotation is proposed to avoid the high-complexity calculation of the cross-correlation matrix of the input sequence in the RLS algorithm, making the DCD algorithm compatible with the nonshifted structure of input sequence in UWA channels.

Second, we propose a low-complexity RLS algorithm based on FDE. We extend the traditional linear channel shortening method to a nonlinear frequency-domain decision-feedback equalization (FD-DFE) to minimize the ISI, consequently yielding an RLS algorithm with a reduced filter length. The overlapping partitioning and weighting processing addresses the mismatch between the blockwise FD-DFE and symbolwise RLS-DFE.

Third, the numerical simulations and experimental results provide a reference for the low-complexity RLS algorithms in different configurations and environments. Both the algorithms are adopted in DFE and examined in the "mild" and "wild" UWA channel conditions and show their superiority over traditional RLS-based DFEs.

The rest of this article is organized as follows. Section II introduces the system model used in the two strategies. Sections III and IV detail the *i*DCD-RLS- and FDE-RLS-based DFEs, respectively. Simulation and experimental results are provided in Sections V and VI, respectively. Finally, Section VII concludes this article.

In the following sections, bold capital letters represent matrices, while bold lowercase letters represent vectors. $\boldsymbol{A}(1,:)$ and $\boldsymbol{A}(:,1)$ denote the first row and first column of matrix $\boldsymbol{A}$, respectively. The ith element of the vector $\mathbf{a}$ is denoted as $a(i)$. $\mathbb{C}$ denotes the complex set. The superscripts $(\cdot)^T$ and $(\cdot)^H$ represent the transpose and Hermitian transpose, respectively. The Hadamard product is represented by $\odot$, and the real and imaginary parts are represented by $\Re(\cdot)$ and $\Im(\cdot)$, respectively. $\angle a$ represents the phase angle of a.

II. System Model

We consider the signal in one packet of a single-input and multiple-output (SIMO) UWA communication system, as shown in Fig. 1. At the transmitter, the information bits are converted to symbols by encoding, interleaving, and mapping. The training

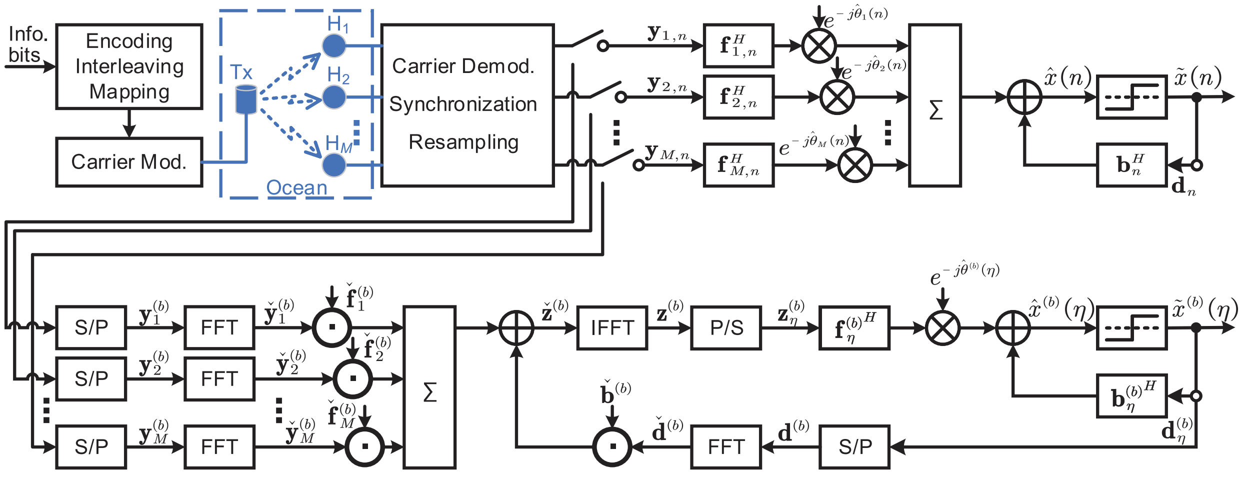


Fig. 1. System model for SIMO UWA communications. $\check{(\cdot)}$ denotes the frequency-domain variable, S/P = serial-parallel converter, P/S = parallel–serial converter, FFT = fast Fourier transform, and IFFT = inverse FFT. Empty circles represent collecting symbols into vectors.

symbols and the synchronization signal are then inserted before the information symbols. At the receiver, the baseband signal is first compensated if the wideband Doppler cannot be neglected. The received signal sampled on the mth hydrophone at instant n can be expressed as

$$y_m(n) = \sum_{l=0}^{L-1} h_m(n,l)x(n-l) + v_m(n) \quad (1)$$

where $x(n)$ is the transmitted symbol, $h_m(n,l)$ is the lth ($l = 0,\dots,L-1$) path gain at instant n between the source and the mth hydrophone, and $v_m(n)$ is the additive white Gaussian noise. Note that the phase rotation is included in the channel gain for the sake of simplicity. Either symbolwise or blockwise equalization can be adopted. In the symbolwise equalization, to mitigate the ISI, a DFE model is adopted

$$\hat{x}(n) = \sum_{m=1}^{M} e^{-j\hat{\theta}_m(n)} \mathbf{f}_{m,n}^H \mathbf{y}_{m,n} + \mathbf{b}_n^H \mathbf{d}_n \quad (2)$$

where M is the number of hydrophones, $\hat{x}(n)$ is the equalized symbol, $\hat{\theta}_m(n)$ denotes the output of the phase-locked loop (PLL) [18], $\mathbf{f}_{m,n} \in \mathbb{C}^{(N_1+N_2+1)\times 1}$ and $\mathbf{b}_n \in \mathbb{C}^{N_3\times 1}$, respectively, denote the feedforward and feedback filter coefficients, $\mathbf{y}_{m,n} = [y_m(n+N_2),\dots,y_m(n-N_1)]^T$ collects the received samples from the mth hydrophone, and $\mathbf{d}_n = [\tilde{x}(n-1),\dots,\tilde{x}(n-N_3)]^T$ collects the decision symbols.

Since the channels within consecutive symbols do not change much, (1) can be written in a block manner as

$$y_m^{(b)}(n) = \sum_{l=0}^{L-1} h_m^{(b)}(l)x^{(b)}(n-l) + v_m^{(b)}(n) \quad (3)$$

where the superscript (b) designates the subblock index, and the channel in the bth subblock $h_m^{(b)}(l)$ is treated as time invariant. Define

$$\mathbf{x}^{(b)} = [x^{(b)}(0),\dots,x^{(b)}(N_b-1)]^T$$
$$\mathbf{y}_m^{(b)} = [y_m^{(b)}(0),\dots,y_m^{(b)}(N_b-1)]^T$$

and

$$\mathbf{v}_m^{(b)} = [v_m^{(b)}(0),\dots,v_m^{(b)}(N_b-1)]^T$$

with N_b being the subblock length; then, (3) can be rewritten as

$$\mathbf{y}_m^{(b)} = \mathbf{H}_m^{(b)}\mathbf{x}^{(b)} + \bar{\mathbf{H}}_m^{(b-1)}\bar{\mathbf{x}}^{(b-1)} + \mathbf{v}_m^{(b)}. \quad (4)$$

$\mathbf{H}_m^{(b)}$ is an $N_b \times N_b$ circulant matrix with the first column being

$$\left[h_m^{(b)}(0),\dots,h_m^{(b)}(L-1),0,\dots,0\right]^T$$

$\bar{\mathbf{H}}_m^{(b-1)}$ is an $N_b \times (L-1)$ Toeplitz matrix with the first row being

$$\left[h_m^{(b-1)}(L-1) - h_m^{(b)}(L-1),\dots,h_m^{(b-1)}(1) - h_m^{(b)}(1)\right]$$

and the first column being

$$\left[h_m^{(b-1)}(L-1) - h_m^{(b-1)}(L-1),0,\dots,0\right]^T$$

and

$$\bar{\mathbf{x}}^{(b-1)} = [x^{(b)}(-L+1) - x^{(b)}(N_b-L+1),\dots,$$
$$\times\, x^{(b)}(-1) - x^{(b)}(N_b-1)]^T.$$

The negative indices indicate the symbols preceding $x^{(b)}(0)$. Applying the $N_b \times N_b$ Fourier transform matrix $\mathbf{F}$ to (4) yields

$$\check{\mathbf{y}}_m^{(b)} = \check{\mathbf{H}}_m^{(b)}\check{\mathbf{x}}^{(b)} + \check{\bar{\mathbf{H}}}_m^{(b-1)}\check{\bar{\mathbf{x}}}^{(b-1)} + \check{\mathbf{v}}_m^{(b)} \quad (5)$$

where $\check{\mathbf{H}}_m^{(b)} = \mathbf{F}\mathbf{H}_m^{(b)}\mathbf{F}^H$ is a diagonal matrix, $\check{\bar{\mathbf{H}}}_m^{(b-1)} = \mathbf{F}\bar{\mathbf{H}}_m^{(b-1)}\mathbf{F}^H(:,1:L-1)$ is an off-diagonal matrix, and the other vectors $\check{(\cdot)}$ denote the corresponding frequency-domain expressions. The second term of (5) represents the intersubblock interference. It can be mitigated by discarding the initial segment after FDE and inverse Fourier transform, since the interference is located at the beginning, as shown in (4). During FDE, a nonlinear preprocessing filter can be exploited for channel shortening

$$\check{\mathbf{z}}^{(b)} = \sum_{m=1}^{M} \check{\mathbf{f}}_m^{(b)} \odot \check{\mathbf{y}}_m^{(b)} + \check{\mathbf{b}}^{(b)} \odot \check{\mathbf{d}}^{(b)} \quad (6)$$

where $\check{\mathbf{z}}^{(b)}$ is the channel-shortened subblock, $\check{\mathbf{f}}_m^{(b)}$ and $\check{\mathbf{b}}^{(b)}$, respectively, denote the feedforward and feedback filter coefficients, which are often calculated by the channel estimates, and $\check{\mathbf{d}}^{(b)}$ denotes the decision subblock. After inverse Fourier transform, a single-channel DFE model is adopted

$$\hat{x}^{(b)}(\eta) = e^{-j\hat{\theta}^{(b)}(\eta)} \mathbf{f}_\eta^{(b)H} \mathbf{z}_\eta^{(b)} + \mathbf{b}_\eta^{(b)H} \mathbf{d}_\eta^{(b)} \tag{7}$$

where η $(\eta = 0, \ldots, N_b - 1)$ is the symbol index in the bth subblock, $\hat{x}^{(b)}(\eta)$ is the equalized symbol, $\mathbf{f}_\eta^{(b)} \in \mathbb{C}^{(N_1'+N_2'+1)\times 1}$ and $\mathbf{b}_\eta^{(b)} \in \mathbb{C}^{N_3'\times 1}$ denote the feedforward and feedback filter coefficients, respectively, and

$$\mathbf{z}_\eta^{(b)} = [z^{(b)}(\eta + N_2'), \ldots, z^{(b)}(\eta - N_1')]^T$$

and $\mathbf{d}_\eta^{(b)} = [\tilde{x}^{(b)}(\eta - 1), \ldots, \tilde{x}^{(b)}(\eta - N_3')]^T$ collect the channel-shortened samples and decision symbols, respectively. The phase estimate $\hat{\theta}^{(b)}(\eta)$ can be calculated by

$$\hat{\theta}^{(b)}(\eta) = \hat{\theta}(n) - \hat{\theta}^{(b)}(0) \tag{8}$$

where $\hat{\theta}(n)$ is the PLL output of the channel-shortened signal [18], and the phase estimate of the first symbol in the subblock $\hat{\theta}^{(b)}(0)$ can be directly obtained from $\hat{\theta}(n)$. The filter lengths $N_i' < N_i$ $(i = 1, \ldots, 3)$ because of channel shortening.

The feedforward and feedback filter coefficients, which can be organized into a vector

$$\hat{\mathbf{w}}_n = [\mathbf{f}_{1,n}^T, \ldots, \mathbf{f}_{M,n}^T, \mathbf{b}_n^T]^T$$

for symbolwise equalization, or

$$\hat{\mathbf{w}}_\eta^{(b)} = [\mathbf{f}_\eta^{(b)T}, \mathbf{b}_\eta^{(b)T}]^T$$

for blockwise equalization, are updated using the RLS-based algorithm during equalization. The details of equalization will be presented in the following two sections.

III. Improved DCD-RLS-Based DFE

In this section, the preliminaries in the DCD-RLS algorithm are first introduced. The improved counterpart for DFE-based equalization is then presented, followed by the complexity analysis.

A. Preliminaries in the DCD-RLS Algorithm

In the RLS algorithm, the cost function consists of two components, i.e., the sum of weighted error squares and the regularizing term

$$J(n) = \sum_{i=1}^{n} \lambda^{n-i} \left| x(n) - \mathbf{w}_n^H \mathbf{u}_i \right|^2 + \delta\lambda^n \left\| \mathbf{w}_n \right\|^2 \tag{9}$$

where λ is the forgetting factor, $x(n)$ is the desired symbol, and $\mathbf{u}_i = [u(i), u(i-1), \ldots, u(i-N+1)]^T$ collects the phase-compensated received symbols and decision symbols. Setting the gradient of $J(n)$ to zero with respect to $\mathbf{w}$ yields the normal equation

$$\mathbf{\Phi}_n \mathbf{w}_n = \mathbf{c}_n \tag{10a}$$

where

$$\mathbf{\Phi}_n = \sum_{i=1}^{n} \lambda^{n-i} \mathbf{u}_i \mathbf{u}_i^H + \delta\lambda^n \mathbf{I} \tag{10b}$$

denotes the time-average correlation matrix of the input vector $\mathbf{u}$ and

$$\mathbf{c}_n = \sum_{i=1}^{n} \lambda^{n-i} \mathbf{u}_i x^*(i) \tag{10c}$$

denotes the time-average cross-correlation vector between the desired signal and $\mathbf{u}$.

In addition to the normal equation of the RLS algorithm, the DCD-RLS algorithm uses an auxiliary normal equation to find the approximate solution $\hat{\mathbf{w}}_n$ [6]

$$\mathbf{\Phi}_n \hat{\mathbf{w}}_n = \hat{\mathbf{c}}_n. \tag{11}$$

In these normal equations, $\mathbf{\Phi}_n$ and $\mathbf{c}_n$ can be written as recursions

$$\mathbf{\Phi}_n = \lambda \mathbf{\Phi}_{n-1} + \mathbf{u}_n \mathbf{u}_n^H \tag{12a}$$

$$\mathbf{c}_n = \lambda \mathbf{c}_{n-1} + x^*(n) \mathbf{u}_n. \tag{12b}$$

Define a residual vector for those two normal equations as

$$\mathbf{a}_{n|n'} = \mathbf{c}_n - \mathbf{\Phi}_n \hat{\mathbf{w}}_{n'} \tag{13}$$

with the subscript n' being n or $n-1$ to enable recursive derivation, and increments of $\mathbf{c}$ and $\mathbf{\Phi}$ as

$$\Delta \mathbf{c}_n = \mathbf{c}_n - \mathbf{c}_{n-1} \tag{14}$$

$$\Delta \mathbf{\Phi}_n = \mathbf{\Phi}_n - \mathbf{\Phi}_{n-1}. \tag{15}$$

If we look into (10a) and (13) (with n' being $n-1$), we have

$$\mathbf{a}_{n|n-1} = \mathbf{\Phi}_n \Delta \mathbf{w}_n \tag{16}$$

where $\Delta \mathbf{w}_n = \mathbf{w}_n - \hat{\mathbf{w}}_{n-1}$. In this way, solving the original problem (10a) with respect to $\mathbf{w}_n$ transforms to dealing with the auxiliary (16) with respect to $\Delta \mathbf{w}_n$. The problem turns to finding the solution $\Delta \mathbf{w}_n$ and calculating the residual vector $\mathbf{a}_{n|n'}$ recursively. The derivation from $\mathbf{a}_{n-1|n-2}$ to $\mathbf{a}_{n|n-1}$ is presented as follows.

Substituting (14) into (13) (with n and n' both being $n-2$) yields

$$\mathbf{a}_{n-2|n-2} = \mathbf{c}_{n-1} - \Delta \mathbf{c}_{n-1} - \mathbf{\Phi}_{n-2} \hat{\mathbf{w}}_{n-2}. \tag{17}$$

Substituting (17) into (13) (with n being $n-1$ and n' being $n-2$) yields

$$\mathbf{a}_{n-1|n-2} = \mathbf{c}_{n-1} - \mathbf{\Phi}_{n-1} \hat{\mathbf{w}}_{n-2} \tag{18a}$$

$$= \mathbf{a}_{n-2|n-2} + \Delta \mathbf{c}_{n-1} - \Delta \mathbf{\Phi}_{n-1} \hat{\mathbf{w}}_{n-2}. \tag{18b}$$

Substituting (18a) into (13) (with n and n' both being $n-1$) yields

$$\mathbf{a}_{n-1|n-1} = \mathbf{a}_{n-1|n-2} - \mathbf{\Phi}_{n-1} \Delta \hat{\mathbf{w}}_{n-1} \tag{19}$$

Algorithm 1: DCD-RLS Algorithm for Shift-Structured Input.

Input: $\mathbf{u}_n$, $\mathbf{\Phi}_{n-1}$, $\hat{\mathbf{w}}_{n-1}$, $\mathbf{a}_{n-1|n-1}$, M_{w}, N_{s}

Output: $\hat{x}(n)$, $\mathbf{\Phi}_n$, $\hat{\mathbf{w}}_n$, $\mathbf{a}_{n|n}$

1 $\hat{x}(n) = \hat{\mathbf{w}}_{n-1}^H \mathbf{u}_n$;

2 $\xi(n) = \tilde{x}(n) - \hat{x}(n)$;

3 **Update** $\mathbf{\Phi}_n$ **by** (21);

4 $\mathbf{a}_{n|n-1} = (1 - 2^{-q})\mathbf{a}_{n-1|n-1} + \xi^*(n)\mathbf{u}_n$;

5 DCD initialization: $\mathbf{w} = \hat{\mathbf{w}}_{n-1}$, $\mathbf{a} = \mathbf{a}_{n|n-1}$, $\boldsymbol{\beta}_0 = [1, -1, j, -j]^T$, $H = 1$, $\boldsymbol{\beta} = H\boldsymbol{\beta}_0$, $m_{\mathrm{w}} = 0$, $i_{\mathrm{s}} = 0$;

6 **while** $m_{\mathrm{w}} < M_{\mathrm{w}}$ & $i_{\mathrm{s}} < N_{\mathrm{s}}$ **do**

7 $\quad [r_0, c_0] = \arg\min_{r,c} (H/2)\Phi_n(r,r) - \Re\{a(r)\beta_0^*(c)\}$;

8 $\quad \Delta J_0 = (H/2)\Phi_n(r_0, r_0) - \Re\{a(r_0)\beta_0^*(c_0)\}$;

9 $\quad$ **if** $\Delta J_0 < 0$ *or* $\Delta J_0 < \Delta J_0^{\min}$ *(*$i_{\mathrm{s}} \geqslant 1$*)* **then**

10 $\quad\quad w(r_0) = w(r_0) + \beta(c_0)$;

11 $\quad\quad \mathbf{a} = \mathbf{a} - \beta(c_0)\mathbf{\Phi}_n(:, r_0)$;

12 $\quad\quad i_{\mathrm{s}} = i_{\mathrm{s}} + 1$;

13 $\quad$ **else if** $\Delta J_0 > 0$ *or* $\Delta J_0 \geqslant \Delta J_0^{\min}$ *(*$i_{\mathrm{s}} \geqslant 1$*)* **then**

14 $\quad\quad \boldsymbol{\beta} = \boldsymbol{\beta}/2$, $H = H/2$, $m_{\mathrm{w}} = m_{\mathrm{w}} + 1$;

15 $\quad$ **end**

16 $\quad \Delta J_0^{\min} = \Delta J_0$;

17 **end**

18 $\hat{\mathbf{w}}_n = \mathbf{w}$, $\mathbf{a}_{n|n} = \mathbf{a}$;

where $\Delta\hat{\mathbf{w}}_{n-1} = \hat{\mathbf{w}}_{n-1} - \hat{\mathbf{w}}_{n-2}$. To avoid the calculation of $\Delta\mathbf{c}$ and $\Delta\mathbf{\Phi}$, (18b) can be transformed to (with n being $n+1$)

$$\mathbf{a}_{n|n-1} = \lambda\mathbf{a}_{n-1|n-1} + \xi^*(n)\mathbf{u}_n \tag{20}$$

via (12)–(15) (see the Appendix), with $\xi(n) = x(n) - \hat{\mathbf{w}}_{n-1}^H\mathbf{u}_n$ being the estimation error. Since the ground truth $x(n)$ is often unavailable, it can be replaced with the decision counterpart $\tilde{x}(n)$.

Up to now, the three variables, $\mathbf{\Phi}$, $\Delta\hat{\mathbf{w}}$, and $\mathbf{a}_{n|n'}$, can be recursively calculated by the DCD-RLS algorithm, as shown in Algorithm 1. If the input vector has a shift structure, i.e., $\mathbf{u}_n = [u(n), u(n-1), \ldots, u(n-N+1)]^T$, updating $\mathbf{\Phi}_n$ can be achieved via copying its upper left $(N-1)\times(N-1)$ part to the lower right $(N-1)\times(N-1)$ part

$$\mathbf{\Phi}_n(2:\mathrm{end}, 2:\mathrm{end}) = \lambda\mathbf{\Phi}_{n-1}(1:\mathrm{end}-1, 1:\mathrm{end}-1) \tag{21a}$$

and calculating the first column and row by

$$\mathbf{\Phi}_n(:,1) = \lambda\mathbf{\Phi}_{n-1}(:,1) + u^*(n)\mathbf{u}_n \tag{21b}$$

$$\mathbf{\Phi}_n(1,:) = \mathbf{\Phi}_n^H(:,1). \tag{21c}$$

$\Delta\hat{\mathbf{w}}$ can be obtained through the coordinate descent algorithm, which updates a single tap using a step at every iteration. In the DCD iterations, the step is replaced by a coordinate $\boldsymbol{\beta}$, which is represented by M_{w} bits within an amplitude interval $[-H, H]$. The algorithm starts updating the most significant bits of the solution and proceeds toward less significant bits while halving $\boldsymbol{\beta}$, as shown in lines 9–15 of Algorithm 1. The maximum number of *successful* iterations is N_{s}, where the cost function ΔJ_0 [8] is negative or smaller than the cost in the previous iteration. The update of $\mathbf{a}_{n|n-1}$ from $\mathbf{a}_{n-1|n-2}$ is achieved by (19) and (20), as illustrated in lines 11 and 4, respectively. It is worth noting that the forgetting factor in (20) is chosen as $\lambda = 1 - 2^{-q}$ with a positive integer q, and H is chosen as a power-of-2 number. Both the choices enable multiplication to be replaced by addition and bit-shift operations.

The DCD-RLS algorithm requires a time-shifted structure of the input vector $\mathbf{u}$ to achieve a fast updating of $\mathbf{\Phi}_n$ by performing the memory address modification [6]. However, in this research, $\mathbf{u}_n = [\mathbf{y}_{1,n}^T e^{-j\hat{\theta}_1(n)}, \ldots, \mathbf{y}_{M,n}^T e^{-j\hat{\theta}_M(n)}, \mathbf{d}_n^T]^T$ is not time shifted due to the fluctuating phase increments in SIMO systems. Albeit being employed in UWA equalization, the complexity of the DCD-RLS algorithm is still $\mathcal{O}[N_0^2]$ [15], [19] as the updating of $\mathbf{\Phi}_n$ dominates the complexity. Hence, the DCD-RLS algorithm should be improved to accommodate to the UWA environment.

B. iDCD-RLS-Based DFE

The autocorrelation matrix of the input vector is

$$\mathbf{\Phi}_n^u = \mathbf{u}_n \mathbf{u}_n^H$$

$$\triangleq \begin{bmatrix} \mathbf{\Phi}_{11,n}^{yy} & \cdots & \mathbf{\Phi}_{1M,n}^{yy} & \mathbf{\Phi}_{1,n}^{dy} \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{\Phi}_{M1,n}^{yy} & \cdots & \mathbf{\Phi}_{MM,n}^{yy} & \mathbf{\Phi}_{M,n}^{dy} \\ \mathbf{\Phi}_{1,n}^{yd} & \cdots & \mathbf{\Phi}_{M,n}^{yd} & \mathbf{\Phi}_{n}^{dd} \end{bmatrix} \tag{22a}$$

with

$$\begin{cases} \mathbf{\Phi}_{mm',n}^{yy} = \mathbf{y}_{m,n}\mathbf{y}_{m',n}^H e^{-j[\hat{\theta}_m(n)-\hat{\theta}_{m'}(n)]} \\ \mathbf{\Phi}_{m,n}^{yd} = \mathbf{d}_n \mathbf{y}_{m,n}^H e^{j\hat{\theta}_m(n)} \\ \mathbf{\Phi}_{m,n}^{dy} = \mathbf{y}_{m,n}\mathbf{d}_n^H e^{-j\hat{\theta}_m(n)} \\ \mathbf{\Phi}_n^{dd} = \mathbf{d}_n \mathbf{d}_n^H \\ 1 \leqslant m, m' \leqslant M. \end{cases} \tag{22b}$$

$\mathbf{\Phi}_n^u$ is a Hermitian matrix, so only the upper or lower triangular submatrices of $\mathbf{\Phi}_n^u$ should be considered. If $m = m'$, $\mathbf{\Phi}_{mm',n}^{yy}$ is independent of $\hat{\theta}$. The same is true for $\mathbf{\Phi}_n^{dd}$. The aforementioned fast updating method for time-shifted input can be adopted here. For other cases, the fast updating method can also be adopted if the phase increments $\Delta\hat{\theta}_m(n) = \hat{\theta}_m(n) - \hat{\theta}_m(n-1)$ and $\Delta\hat{\theta}_{mm'}(n) = \Delta\hat{\theta}_m(n) - \Delta\hat{\theta}_{m'}(n)$ are small enough. The only part for updating is the first row and the first column, which is accomplished by

$$\mathbf{\Phi}_{mm',n}^{yy}(:,1) = \lambda\mathbf{\Phi}_{mm',n-1}^{yy}(:,1) + y_{m'}^*(n+N_2)\mathbf{y}_{m,n} \tag{23a}$$

$$\mathbf{\Phi}_{mm',n}^{yy}(1,:) = \begin{cases} \left[\mathbf{\Phi}_{mm',n}^{yy}(:,1)\right]^H, & m = m' \\ \lambda\mathbf{\Phi}_{mm',n-1}^{yy}(1,:) \\ \quad +y_m(n+N_2)\mathbf{y}_{m',n}^H, & m \neq m' \end{cases} \tag{23b}$$

for $\mathbf{\Phi}_{mm',n}^{yy}$ and by

$$\mathbf{\Phi}_n^{dd}(:,1) = \lambda\mathbf{\Phi}_{n-1}^{dd}(:,1) + \tilde{x}^*(n)\mathbf{y}_{m,n} \tag{24a}$$

$$\mathbf{\Phi}_n^{dd}(1,:) = \left[\mathbf{\Phi}_n^{dd}(:,1)\right]^H \tag{24b}$$

for $\mathbf{\Phi}_n^{dd}$.

When the phase variation is significant and the filter length is long, the accumulated phase increments cannot be neglected. The submatrices of $\mathbf{\Phi}_{mm',n}^{yy}$ $(m \neq m')$ and $\mathbf{\Phi}_{m,n}^{dy}$ (or $\mathbf{\Phi}_{m,n}^{yd}$) should be further corrected. Assume that $N_1 + N_2 + 1 \geqslant N_3$, and define the accumulated phase increments

$$\begin{aligned} \Theta_m(n) &= \sum_{i=1}^{N_3-1} \Delta\hat{\theta}_m(n-i+1) \\ &= \hat{\theta}_m(n) - \hat{\theta}_m(n-N_3+1) \end{aligned} \tag{25a}$$

$$\begin{aligned} \Theta_{mm'}(n) &= \sum_{i=1}^{N_1+N_2} \Delta\hat{\theta}_{mm'}(n-i+1) \\ &= \left[\hat{\theta}_m(n) - \hat{\theta}_m(n-N_1-N_2)\right] \\ &\quad - \left[\hat{\theta}_{m'}(n) - \hat{\theta}_{m'}(n-N_1-N_2)\right]. \end{aligned} \tag{25b}$$

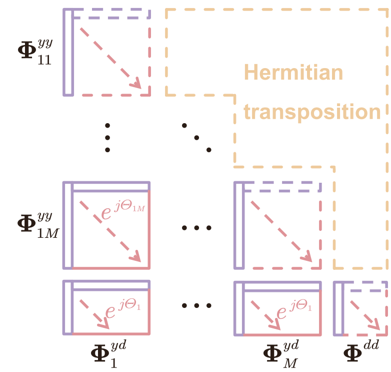


Fig. 2. Updating of the correlation matrix. Solid lines represent subvectors or submatrices being updated with multiplications, while dash lines represent those without multiplications. The first rows of diagonal submatrices can be obtained by taking the Hermitian transpose of the first column, and the remaining part can be updated by memory address modification. The first columns and rows of the off-diagonal submatrices should be calculated by (23), and the remaining part should be phase corrected by (26) only when the accumulated phase increments exceed the threshold.

If $\left|\Theta_{mm'}(n)\right|$ and $\left|\Theta_m(n)\right|$ exceed a threshold Θ_{th}, the lower right parts of $\mathbf{\Phi}_{mm',n}^{yy}$ and $\mathbf{\Phi}_{mn}^{yd}$ should be corrected according to the accumulated phase increments, i.e.,

$$\mathbf{\Phi}_{mm',n}^{yy}(2:\text{end}, 2:\text{end}) = e^{j\Theta_{mm'}(n)}\mathbf{\Phi}_{mm',n}^{yy}(2:\text{end}, 2:\text{end}) \tag{26a}$$

$$\mathbf{\Phi}_{m,n}^{yd}(2:\text{end}, 2:\text{end}) = e^{j\Theta_m(n)}\mathbf{\Phi}_{m,n}^{yd}(2:\text{end}, 2:\text{end}). \tag{26b}$$

If the threshold Θ_{th} is large, the phase correction is rarely performed but with critical errors in updating the correlation matrix; if the threshold is small, the updating of correlation matrix would be precise but with a large computational overhead. Hence, the threshold serves for the tradeoff between the performance and complexity. Finally, the updating of correlation matrix is depicted in Fig. 2. The subscript n is dropped here for the purpose of simplicity. The overall complexity of correlation matrix updating remains low as the constant phase compensation of the full-scale submatrix is avoided.

The *i*DCD-RLS algorithm is shown in Algorithm 2, which differs from Algorithm 1 in equalization and updating of $\mathbf{\Phi}$. After the DFE and phase update (lines 1–4) for instant n, $\mathbf{\Phi}_n$ is updated by the input vector and the accumulated phase increments, as shown in lines 5–15. The obtained $\mathbf{\Phi}_n$ is then used in the improved DCD algorithm to update $\mathbf{w}_n$, as shown in lines 18–29. In the DCD iteration, the weight increment $\Delta\hat{\mathbf{w}}$ and the residual vector $\mathbf{a}_{n|n'}$ are computed with low complexity and assist in calculating the cost function ΔJ_0 in line 20.

C. Complexity

According to Algorithm 2, the complexity of *i*DCD-RLS-based DFE is listed in Table I. The complex multiplications (CMs) and complex additions (CAs) are counted for the equalization of each symbol. In updating the correlation matrix, updating the first rows and columns dominates the complexity, as shown in line 6 of Algorithm 2. Note that the phase correction in lines 10–12 does not cost much computational overhead indeed, as the accumulated phase increments seldom exceed the

Algorithm 2: *i*DCD-RLS Algorithm for DFE-Based Equalization.

Input: $\boldsymbol{\Phi}_{n-1}$, $\hat{\mathbf{w}}_{n-1}$, $\mathbf{a}_{n-1|n-1}$, M_{w}, N_{s}, $\theta_m(0) = 0$

Output: $\hat{x}(n)$, $\boldsymbol{\Phi}_n$, $\hat{\mathbf{w}}_n$, $\mathbf{a}_{n|n}$

1 $\hat{x}^{\mathrm{ff}}_m(n) = e^{-j\hat{\theta}_m(n)}\mathbf{f}^H_{m,n-1}\mathbf{y}_{m,n}$, $\hat{x}^{\mathrm{fb}}(n) = \mathbf{b}^H_{n-1}\mathbf{d}_n$;

2 $\hat{x}(n) = \sum_{m=1}^{M} \hat{x}^{\mathrm{ff}}_m(n) + \hat{x}^{\mathrm{fb}}(n)$;

3 $\xi(n) = \tilde{x}(n) - \hat{x}(n)$;

4 $\varphi_m(n) = \Im\left\{\hat{x}^{\mathrm{ff}}_m(n)\left[\hat{x}^{\mathrm{ff}}_m(n) + \xi(n)\right]^*\right\}$, $\hat{\theta}_m(n+1) = \hat{\theta}_m(n) + K_{f1}\varphi_m(n) + K_{f2}\sum_{n'=0}^{n}\varphi_m(n')$;

5 **Update** $\boldsymbol{\Phi}^{yy}_{mm',n}$ *(*$m \geqslant m'$*)*, $\boldsymbol{\Phi}^{yd}_{m,n}$, and $\boldsymbol{\Phi}^{dd}_n$ **by**

6   Updating the first rows and columns by (23) and (24);

7   Updating the rest part by memory address modification;

8   **if** $m > m'$ **then**

9   Calculate the accumulated phase increment by (25);

10   **if** $\left|\Theta_{mm'}(n)\right| > \Theta_{\mathrm{th}}$ *(or* $\left|\Theta_m(n)\right| > \Theta_{\mathrm{th}}$*)* **then**

11   Conduct phase correction by (26);

12   **end**

13   **end**

14 **end**

15 **Update** $\boldsymbol{\Phi}^{yy}_{mm',n}$ $(m < m')$ and $\boldsymbol{\Phi}^{dy}_{m,n}$ **by** Hermitian transpose;

16 $\mathbf{a}_{n|n-1} = (1 - 2^{-q})\mathbf{a}_{n-1|n-1} + \xi^*(n)\mathbf{u}_n$;

17 DCD initialization: $\mathbf{w} = \hat{\mathbf{w}}_{n-1}$, $\mathbf{a} = \mathbf{a}_{n|n-1}$, $\boldsymbol{\beta}_0 = [1, -1, j, -j]^T$, $H = 1$, $\boldsymbol{\beta} = H\boldsymbol{\beta}_0$, $m_{\mathrm{w}} = 0$, $i_{\mathrm{s}} = 0$;

18 **while** $m_{\mathrm{w}} < M_{\mathrm{w}}$ & $i_{\mathrm{s}} < N_{\mathrm{s}}$ **do**

19     $[r_0, c_0] = \arg\min_{r,c}(H/2)\mathit{\Phi}_n(r,r) - \Re\{a(r)\beta^*_0(c)\}$;

20     $\Delta J_0 = (H/2)\Phi_n(r_0, r_0) - \Re\{a(r_0)\beta^*_0(c_0)\}$;

21   **if** $\Delta J_0 < 0$ *or* $\Delta J_0 < \Delta J^{\min}_0$ *(*$i_{\mathrm{s}} \geqslant 1$*)* **then**

22         $w(r_0) = w(r_0) + \beta(c_0)$;

23         $\mathbf{a} = \mathbf{a} - \beta(c_0)\boldsymbol{\Phi}_n(:, r_0)$;

24         $i_{\mathrm{s}} = i_{\mathrm{s}} + 1$;

25   **else if** $\Delta J_0 > 0$ *or* $\Delta J_0 \geqslant \Delta J^{\min}_0$ *(*$i_{\mathrm{s}} \geqslant 1$*)* **then**

26         $\boldsymbol{\beta} = \boldsymbol{\beta}/2$, $H = H/2$, $m_{\mathrm{w}} = m_{\mathrm{w}} + 1$;

27   **end**

28     $\Delta J^{\min}_0 = \Delta J_0$;

29 **end**

30 $\hat{\mathbf{w}}_n = \mathbf{w}$, $\mathbf{a}_{n|n} = \mathbf{a}$;

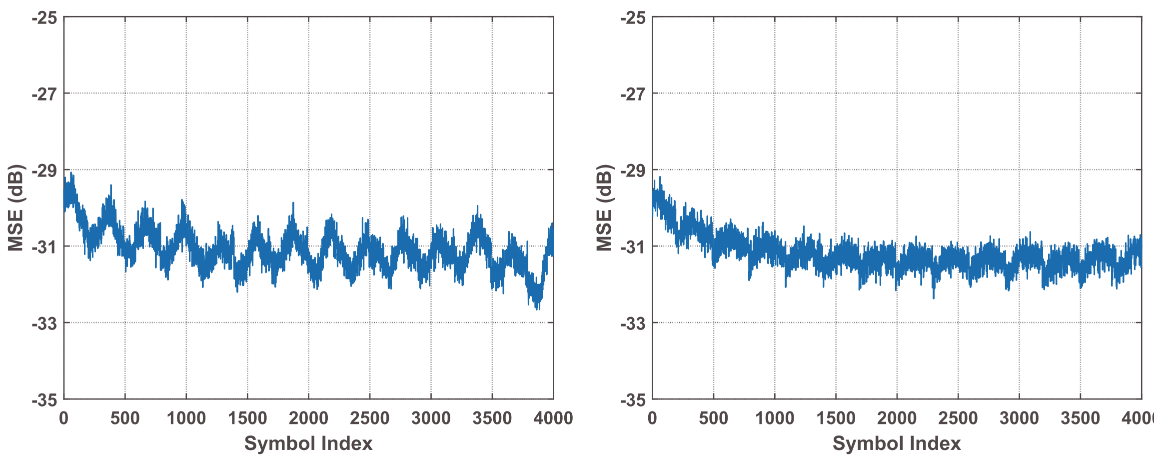
threshold within the filter length. In the DCD algorithm, due to the forgetting factor, which is $\lambda = 1 - 2^{-q}$, most multiplications are replaced by bit shifts and additions. In line 19, finding the minimum is equivalent to finding the maximum, i.e.,

$$[r_0, c_0] = \arg\max_{r,c} \Re\{a(r)\beta^*_0(c)\} \tag{27}$$

which can be achieved by comparing the real and imaginary parts of $a(r)$. Hence, the operation in Line 19 costs $(N_1 + N_2 + 1) + N_3$ CAs $(2(N_1 + N_2 + 1) + 2N_3$ real-valued additions). The complexity of DCD algorithm is limited by the number of *successful* iterations N_{s}. In the worst case, $M_{\mathrm{w}} - 1$ *unsuccessful* updates are performed, and the number

TABLE I
COMPLEXITY OF *i*DCD-RLS-BASED DFE

Lines of Algorithm 2	CMs	CAs
1–3	$M(N_1+N_2+2)+N_3$	$M(N_1+N_2+1)+N_3$
4	$3M$	$4M$
6	$(M+1)\left[M(N_1+N_2+1/2)+N_3\right]$	$2(M+1)\left[M(N_1+N_2+1/2)+N_3\right]$
9	0	$M(3M-1)/2$
10–12	$\leqslant \frac{M^2-M}{2}(N_1+N_2)^2+M(N_1+N_2)(N_3-1)$	0
16	$M(N_1+N_2+1)+N_3$	$2M(N_1+N_2+1)+2N_3$
19	0	$M(N_1+N_2+1)+N_3$
22	0	1
23	0	$M(N_1+N_2+1)+N_3$



(a)

(b)

Fig. 3. MSE performance in FDE-RLS-based DFEs. (a) MSE without using the subblock forgetting factor. (b) MSE with $\mu = 0.3$.

of *successful* updates reaches N_s. The upper limit of DCD algorithm (lines 18–29) is then

$$(M_\mathrm{w}+2N_\mathrm{s})[M(N_1+N_2+1)+N_3]+N_\mathrm{s}.$$

IV. FDE-RLS-BASED DFE

The MMSE-based channel-shortening filter in the frequency domain is essentially a linear FDE filter. It simplifies the followed RLS algorithm as the filter lengths can be much reduced. However, in fast time-varying UWA channels, the channel-shortening ability is limited because of the residual ISI. A heuristic nonlinear FDE filter, namely, the FD-DFE filter, is adopted here to better suppress the residual ISI and shorten the channel. Although the strategy of combining FDE and RLS algorithms is straightforward, the mismatch between the blockwise and symbolwise equalizations should be addressed. The mismatch originates from two operations, i.e., channel estimation and subblock partitioning. In the following section, those two operations are investigated, and the FDE-RLS-based DFE algorithm is formalized, followed by the complexity analysis.

A. FDE-RLS-DFE

Because the channel estimation in one subblock is not constrained by adjacent subblocks, the channel-shortened signals $\check{\mathbf{z}}_b$ among subblocks are inconsistent. In other words, the standalone channel estimate results in the large MSE of RLS-based DFE at the beginning of each subblock, as depicted in Fig. 3(a). To alleviate the inconsistency, we exploit the weighted channel estimates by inducing a subblock forgetting factor for FDE, i.e.,

$$h_m^{(b)}(l) = \sum_{i=1}^{b} \mu(1-\mu)^{b-i}\hat{h}_m^{(i)}(l) \tag{28}$$

where $\hat{h}_m^{(i)}(l)$ denotes the channel estimates in the ith subblock, and $\mu \in (0,1]$ denotes the subblock forgetting factor. Similar to the role of forgetting factor λ in the symbolwise RLS algorithm, the weight assigned to the channel estimate increases as i approaches the current subblock b. After some algebra, (28) can be computed recursively with low complexity

$$h_m^{(b)}(l) = (1-\mu)h_m^{(b-1)}(l) + \mu\hat{h}_m^{(b)}(l). \tag{29}$$

In this way, the subblock inconsistency can be alleviated, as illustrated in Fig. 3(b). After zero padding, $\mathbf{h}_m^{(b)} = [h_m^{(b)}(0),\ldots,h_m^{(b)}(L-1),0,\ldots,0]^T \in \mathbb{C}^{N_b'\times 1}$ (N_b' denoting the length of symbols for FDE or FD-DFE) can be transformed to frequency-domain counterpart $\check{\mathbf{h}}_m^{(b)}$ and used to calculate the filter coefficients for channel shortening.

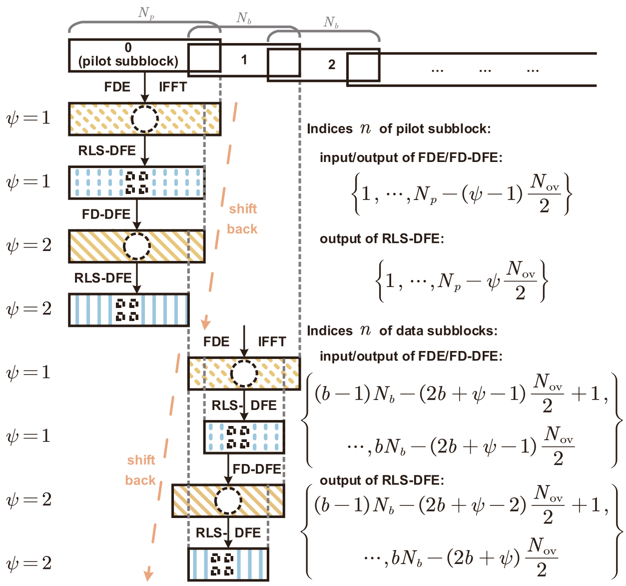


Fig. 4. Subblock structure of the FDE-based RLS algorithm. Two iterations are presented here. The lengths and number of subblocks are also given. b, N_p, ψ, and Ψ denote the subblock index, number of pilot symbols, iteration index, and number of iterations, respectively. $b=0$ designates the pilot subblock, $b \geqslant 1$ designate the data subblocks, and the negative indices of data subblocks correspond to the tail of the pilot subblock.

To address the mismatch in subblock partitioning, we designed an iterative overlapping subblock structure, as shown in Fig. 4. Two iterations are presented here as an example. For the first iteration ($\psi=1$), the linear FDE is conducted as no feedback is provided. For the followed iterations ($\psi>1$), the nonlinear FD-DFE is employed to mitigate the residual ISI. In FD-DFE, the filter coefficients are optimized to minimize the MSE of the equalized symbols

$$\begin{aligned} J &= E\left\{\left|\check{z}^{(b)}(\eta)-\check{x}^{(b)}(\eta)\right|^2\right\} \\ &= E\left\{\left|\sum_{m=1}^{M}\check{f}_m^{(b)}(\eta)\left[\check{h}_m^{(b)}(\eta)\check{x}^{(b)}(\eta)+\check{v}_m^{(b)}(\eta)\right]\right.\right. \\ &\qquad \left.\left.+\check{b}^{(b)}(\eta)\check{\tilde{x}}^{(b)}(\eta)-\check{x}^{(b)}(\eta)\right|^2\right\}. \end{aligned} \tag{30}$$

The frequency-domain feedforward filter coefficient is

$$\check{f}_m^{(b)}(\eta)=\kappa\check{g}_m^{(b)}(\eta) \tag{31a}$$

With

$$\check{g}_m^{(b)}(\eta)=\frac{\left[\check{h}_m^{(b)}(\eta)\right]^*}{\left(1-\varrho_{\psi-1}^2\right)\sum_{m=1}^{M}\left|\check{h}_m^{(b)}(\eta)\right|^2+\sigma^2} \tag{31b}$$

$$\kappa=N_b'\left[\sum_{m=1}^{M}\sum_{\eta=0}^{N_b'-1}\check{g}_m^{(b)}(\eta)\check{h}_m^{(b)}(\eta)\right]^{-1} \tag{31c}$$

where $\check{h}_m^{(b)}(\eta)$ represents the frequency-domain channel estimate, and σ^2 represents the variance of noise. ϱ_ψ is the decision reliability in the ψth iteration, whose calculation is described in [20]. The frequency-domain feedback filter coefficient is

$$\check{b}^{(b)}(\eta)=1-\sum_{m=1}^{M}\check{f}_m^{(b)}(\eta)\check{h}_m^{(b)}(\eta). \tag{32}$$

The channel-shortened signal is then

$$\check{\mathbf{z}}^{(b)}=\sum_{m=1}^{M}\check{\mathbf{f}}_m^{(b)}\odot\check{\mathbf{y}}_m^{(b)}+\check{\mathbf{b}}^{(b)}\odot\check{\tilde{\mathbf{d}}}^{(b)} \tag{33}$$

where

$$\begin{aligned}\check{\mathbf{f}}_m^{(b)} &= [\check{f}_m^{(b)}(0),\ldots,\check{f}_m^{(b)}(N_b'-1)]^T \\ \check{\mathbf{b}}^{(b)} &= [\check{b}^{(b)}(0),\ldots,\check{b}^{(b)}(N_b'-1)]^T\end{aligned}$$

and $\check{\tilde{\mathbf{d}}}^{(b)}$ is the soft decision output in the $(\psi-1)$th iteration. In the first iteration, $\check{\tilde{\mathbf{d}}}^{(b)}=\mathbf{0}_{N_b'\times 1}$. One may refer to [21] and [22] for the details of filter in (33). Since no cyclic prefix or zero padding is inserted in the transmitted sequence, the channel matrix within each subblock is not circulant, which leads to the precursor and postcursor interference in the subblock. The overlapping partitioning [20], [23] is then adopted in the receiver. Among adjacent subblocks, there exist $N_{\rm ov}$ overlapping symbols ($N_{\rm ov}/2$ should be no less than $N_2'-1$), and first $N_{\rm ov}/2$ symbols are discarded to mitigate the precursor interference. Afterward, the groupwise phase estimation [24] is conducted to calculate phase rotation induced by Doppler shift. To make the groupwise phase rotation compatible with the PLL output, the former can be interpolated and yields the symbolwise estimates $\hat{\theta}_D(n)$.

In time-domain RLS-based DFE, the carrier phase distortion should be carefully considered. The phase rotation induced by the Doppler shift should be compensated for, and the input symbols should be phase shifted by the PLL output of the first symbol, $\hat{\theta}^{(b)}(0)$. Hence, the single-channel DFE in (7) should be rewritten as

$$\hat{x}^{(b)}(\eta)=\mathbf{f}_\eta^{(b)H}\left(\boldsymbol{\zeta}\odot\mathbf{z}_\eta^{(b)}\right)+\mathbf{b}_\eta^{(b)H}\mathbf{d}_\eta^{(b)} \tag{34}$$

where the n'th ($n'=1,\ldots,N_1'+N_2'+1$) element of $\boldsymbol{\zeta}$ is

$$\zeta(n')=e^{-j\left[(N_2'-n')\hat{\theta}_D(n)/R_{\rm s}+\hat{\theta}(n)-\hat{\theta}^{(b)}(0)\right]} \tag{35}$$

with $R_{\rm s}$ being the symbol rate. In this way, the phase jump among adjacent subblocks is avoided. Note that the weight of RLS-based DFE should be initialized as a zero vector except for the N_2'th element being one, since the main ISI has been mitigated in channel shortening. At the end of each iteration, as the last $N_{\rm ov}/2$ symbols cannot be recovered and fed to the next iteration, the subblock should shift back $N_{\rm ov}/2$ symbols, as shown in Fig. 4. In each iteration, the equalized signal for RLS-based DFE is consecutive across subblocks.

Up to now, the FDE-RLS algorithm can be summarized in Algorithm 3, where M_b denotes the number of subblocks. Lines 5–9 indicate the process of channel shortening, while lines 10–25 indicate the process of RLS-based DFE. The phase estimation using a combination of the groupwise [24] and symbolwise [18] methods is demonstrated in lines 11–17. The average phase deviation in line 13 is computed for quadrature

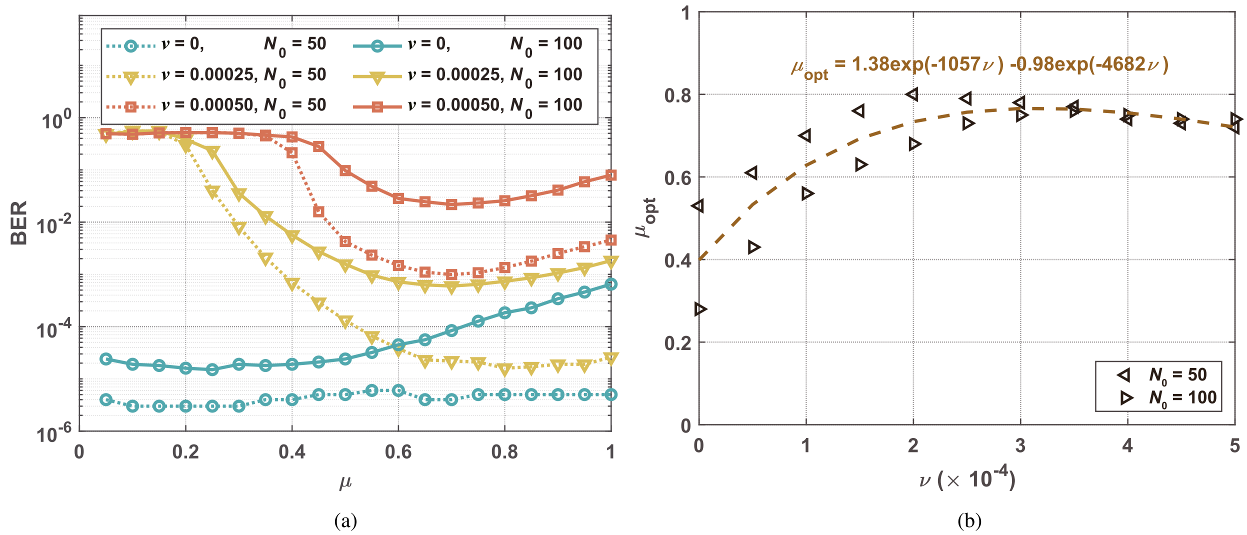


Fig. 5. Analysis of μ considering Doppler shifts. (a) BER-μ performance curves. (b) Optimal μ's for different Doppler shifts.

phase-shift keying (QPSK) modulations, and the second term in the brackets needs to be adjusted for other modulations. It is worth noting that the RLS algorithm in line 23 can be replaced by the proposed *i*DCD-RLS algorithm. However, the complexity reduction is limited as the channel has been shortened and the multichannel signals has been converted to a single composite signal.

B. Choice of μ

In Algorithm 3, the weight μ remains undetermined for the FDE-RLS-based DFE. Intuitively, its optimal value depends on Doppler, as a large μ is required to ensure the real-time channel tracking when Doppler is small. On the contrary, a large Doppler decreases the channel correlation among subblocks and, thus, intensifies the *inconsistency* after channel shortening. As a consequence, the MSE increases at the overlapping parts and the weight μ should be tuned down to alleviate the *inconsistency*. Hence, we first evaluate the impact of μ on the bit error rate (BER) performance for different Doppler shifts.

Uniformly distributed channels are used in the evaluation with $N_0 = 50$ and 100. The channel length after shortening (i.e., N_0') is set as 40,[1] and the normalized Doppler frequencies (i.e., νs) are set as 0, 2.5×10^{-4}, and 5×10^{-4}. At the receiver, the received signal-to-noise ratio (SNR) is 15 dB, the forgetting factor $\lambda = 0.999$, the subblock length $N_b = 500$, and three iterations are conducted within each subblock. μ is set as 0.05–1 with a step of 0.05, and the corresponding BER curves are presented in Fig. 5(a). The trend of optimal weights is consistent with the previous inference, except for $\nu = 0$, where a small weight works for both the short and long channels. In case of $N_0 = 50$, μ has little effect on BER performance since the zero-value Doppler and short channel length produce a time-invariant shortened channel. It becomes time varying for $N_0 = 100$ with fluctuating residual multipaths, and thus, a small μ should be adopted. As a consequence, the optimal μs can be plotted for different νs as in Fig. 5(b) and fitted by

$$\mu_{\text{opt}} = 1.38e^{-1057\nu} - 0.98e^{-4682\nu}. \tag{36}$$

According to [25], the normalized Doppler ν is calculated by the correlation between the channels in bth and $(b-1)$th subblocks, $\rho_{b-1,b}$, as given by

$$\nu = \frac{\sqrt{-\ln(\rho_{b-1,b})/2}}{\pi(N_b - N_{\text{ov}})}. \tag{37}$$

C. Complexity

The complexity of the FDE-RLS-based DFE is shown in Table II, where CMs are counted for each symbol in each iteration. It is observed that the complexity increases with the number of iterations Ψ and the sparsity P (the number of nonzero taps). Channel estimation dominates the computational complexity. Here, we adopt the naive orthogonal matching pursuit (OMP) [26]. The complexity is

$$\mathcal{O}[PLN_b] + \mathcal{O}[P(P+1)(P+2)N_b]$$

for each channel in each iteration. In comparison, the (inverse) Fourier transform, filter design, channel shortening, and phase estimation have low complexities. The following RLS-based DFE is also computationally efficient due to the single composite signal and the short filter length.

V. Simulation Results

In this section, we present the simulation results based on a public acoustic channel simulator [27]. This simulator takes into account the small- and large-scale channel variations based on the notion of the underlying random displacement being on the order of a few or many wavelengths, respectively. The simulator has been validated using real data from several field trials,

[1] Although not plotted here, the length of a shortened channel does not impact the BER performance except for channels with large Doppler shifts, where small N_0' results in residual ISI. After examining the BERs with different N_0's, $N_0' = 40$ appears to be a good choice for different Doppler shifts.

Algorithm 3: FDE-RLS Algorithm for DFE-Based Equalization.

Input: $\mathbf{y}_m = [y_m(0), \cdots, y_m(n), \cdots]^T$ $(m = 1, \cdots, M)$, $\varPsi$, $\theta_D^{(-1,0)} = 0$, $\hat{\theta}(0) = 0$

Output: $\hat{\mathbf{x}} = [\hat{x}(0), \cdots, \hat{x}(n), \cdots]^T$

Initialization: $\mathbf{f}_\eta^{(b)} = \left[\mathbf{0}_{1\times(N_2'-1)}, 1, \mathbf{0}_{1\times(N_1'+1)}\right]^T$, $\mathbf{b}_\eta^{(b)} = \mathbf{0}_{N_3'\times 1}$, $\mathbf{h}_{m,0} = \mathbf{0}_{L\times 1}$;

for $b = 0 : M_b$ **do**

  for $\psi = 1 : \varPsi$ **do**

    Input subblock signals $\mathbf{y}_m^{(b)}$ (with indices n shown in Fig. 4), channel estimates $\hat{h}_m^{(b)}(l)$;

    $\hat{h}_m^{(b)}(l) \xrightarrow{(29)} h_m^{(b)}(l)$;

    $\mathbf{y}_m^{(b)}$, zero-padded $\mathbf{h}_m^{(b)} \xrightarrow{\text{FFT}} \check{\mathbf{y}}_m^{(b)}$, $\check{\mathbf{h}}_m^{(b)}$;

    $\check{\mathbf{h}}_m^{(b)} \xrightarrow{(31),(32)} \check{\mathbf{f}}_m^{(b)}$, $\check{\mathbf{b}}^{(b)}$;

    $\check{\mathbf{f}}_m^{(b)}$, $\check{\mathbf{b}}^{(b)}$, $\check{\mathbf{y}}_m^{(b)}$, $\check{\bar{\mathbf{d}}}^{(b)} \xrightarrow{(33)} \check{\mathbf{z}}^{(b)}$;

    $\check{\mathbf{z}}^{(b)} \xrightarrow[\text{extraction}]{\text{IFFT}} \mathbf{z}^{(b)}$;

    $\mathbf{z}^{(b)} \xrightarrow{\text{partitioning}} M_\mathrm{g}$ subgroups with each subgroup $\mathbf{z}^{(b,m_\mathrm{g})} = \left[z^{(b)}((m_\mathrm{g}-1)N_\mathrm{g}), \cdots, z^{(b)}(m_\mathrm{g}N_\mathrm{g}-1)\right]$ (N_g is the length of each subgroup except that the last subgroup may contain fewer than N_g symbols);

    Phase compensation for $m_\mathrm{g} = 0$: $\theta_D^{(b,0)} = \theta_D^{(b-1,\mathrm{end})}$, $\mathbf{z}^{(b,0)} = e^{-j\theta_D^{(b,0)}}\mathbf{z}^{(b,0)}$;

    for $m_\mathrm{g} = 1 : M_\mathrm{g} - 1$ **do**

      Calculate the average phase deviation: $\Delta\theta_D^{(b,m_\mathrm{g})} = (1/N_\mathrm{g})\sum_{\eta=(m_\mathrm{g}-1)N_\mathrm{g}}^{m_\mathrm{g}N_\mathrm{g}-1}\left[\angle z^{(b,m_\mathrm{g})}(\eta) - \pi\left(2\lceil 2\angle z^{(b,m_\mathrm{g})}(\eta)/\pi\rceil - 1\right)/4\right]$;

      Calculate the group-wise phase rotation: $\theta_D^{(b,m_\mathrm{g})} = \theta_D^{(b,m_\mathrm{g}-1)} + \Delta\theta_D^{(b,m_\mathrm{g})}$;

      Phase compensation for the m_gth subgroup: $\mathbf{z}^{(b,m_\mathrm{g})} = e^{-j\theta_D^{(b,m_\mathrm{g})}}\mathbf{z}^{(b,m_\mathrm{g})}$;

    end

    $\theta_D^{(b,m_\mathrm{g})} \xrightarrow{\text{interpolation}}$ symbol-wise $\hat{\theta}_D(n)$;

    for $\eta = N_\mathrm{ov}/2 : N_b' - N_\mathrm{ov}/2 - 1$ **do**

      $\hat{\theta}_D(n)$, $\hat{\theta}(n) \xrightarrow{(35)} \boldsymbol{\zeta}$;

      $\hat{x}^{(b),\mathrm{ff}} = \mathbf{f}_\eta^{(b)H}\left(\boldsymbol{\zeta} \odot \mathbf{z}_\eta^{(b)}\right)$, $\hat{x}^{(b),\mathrm{fb}} = \mathbf{b}_\eta^{(b)H}\mathbf{d}_\eta^{(b)}$;

      $\hat{x}^{(b)}(\eta) = \hat{x}^{(b),\mathrm{ff}} + \hat{x}^{(b),\mathrm{fb}}$;

      $\xi(n) = \tilde{x}(n) - \hat{x}(n)$;

      Update $\hat{\mathbf{w}}^{(b)} = \left[\mathbf{f}_\eta^{(b)T}, \mathbf{b}_\eta^{(b)T}\right]^T$ **by** RLS algorithm;

      $\varphi(n) = \Im\left\{\hat{x}^{(b),\mathrm{ff}}\left[\hat{x}^{(b),\mathrm{ff}} + \xi(n)\right]^*\right\}$, $\hat{\theta}(n+1) = \hat{\theta}(n) + K_{f1}\varphi(n) + K_{f2}\sum_{n'=0}^{n}\varphi(n')$;

    end

    Shift back $\mathbf{y}_m^{(b)}$ by $N_\mathrm{ov}/2$ as shown in Fig. 4;

  end

end

TABLE II
COMPLEXITY OF FDE-RLS-BASED DFE

Lines of Algorithm 3	Operations	CMs
4–5	Channel Estimation	$M\{\mathcal{O}[PL]+\mathcal{O}[P(P+1)(P+2)]\}$
6	Fourier Transform	$\mathcal{O}[(M+1/2)\log_2 N_b]$
7	Filter Coefficients	$M(3\Psi-1)$
8	Channel Shortening	$(M+1)-1$
9	Inverse Fourier Transform	$\mathcal{O}[(1/2)\log_2 N_b]$
11–17	Phase estimation	$2N_b$
18–25	RLS-based DFE	$N_b+\mathcal{O}\left[(N_1'+N_2'+N_3'+1)^2\right]$

TABLE III
SIMULATION PARAMETERS IN (a) CHANNEL SIMULATOR AND (b) RLS-BASED DFES

(a)

Parameters	Values	Parameters	Values
Water depth	50 m	Variance of small-scale surface variations	1.5
Transmitter depth	10 m	Variance of small-scale bottom variations	0.75
Receiver depths	5 m, 10 m, 15 m, 20 m	Range of surface height variation	[–1 1] m or [–10 10] m
Transmission ranges	500 m, 1000 m	Range of transmitter height variation	[–0.5 0.5] m or [–5 5] m
Spreading factor	1.5	Range of receiver height variation	[–0.5 0.5] m or [–5 5] m
Sound speed in water	1500 m/s	Range of transmission range variation	[–2 2] m or [–20 20] m
Sound speed in bottom	1540 m/s	Standard deviations of large-scale variations	0.1 m or 1 m
Frequency band	10–12 kHz	Maximum drifting speed	0.03 m/s or 0.3 m/s

(b)

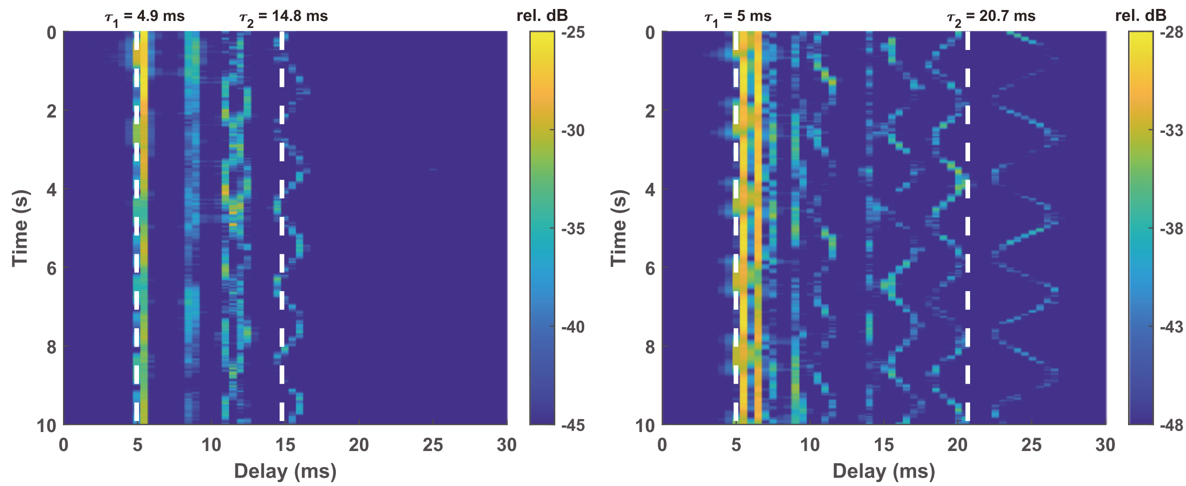
Parameters	Values	Parameters	Values
Duration of training sequence	0.25 s	Ψ	3
λ	$1-2^{-9}$	P	10
N_i $(i=1,2,3)$	[0, 19, 10] or [0, 31, 16]	Duration of channel estimates	0.010 s or 0.016 s
H	1	Block duration	0.06 s
Proportional tracking constant	0.1	Overlapping rate	30%
Integral tracking constant	0.01	N_i' $(i=1,2,3)$	[0, 11, 6] or [0, 18, 9]

such as the Surface Processes and Acoustic Communications Experiment in 2008.

Two sets of channels are generated with the channel simulator, as shown in Table III(a). The water depth is set as 50 m. A transmitter with a frequency band of 10–12 kHz is placed at a depth of 10 m. A four-element receiver array is deployed in the water depth of 5–20 m with an element spacing of 5 m. In the first channel set, the transmission range is 500 m, and the channel delay spread is around 10 ms, as shown in Fig. 6(a). The starting and ending delays of an individual impulse response $h(l)$ are calculated by the adaptive thresholding algorithm in [28]. Those channels are referred to as "mild" channels whose delay spreads and fluctuation rates are small. In the RLS and *i*DCD-RLS algorithms, $\lambda=1-2^{-9}$, $N_1=0$, $N_2=19$, and $N_3=10$; in the FDE-RLS algorithm, the block duration is set as 60 ms, $\lambda=1-2^{-9}$, $N_1'=0$, $N_2'=11$, and $N_3'=6$, as shown in Table III(b). In the second set, the transmission range is 1000 m, leading to a longer delay spread of 16 ms, as shown in Fig. 6(b). The small- and large-scale variations (drifts and deviations of the transmitter and receiver) are larger in the second set, leading to a greater channel fluctuation. Those channels are referred to as "wild" channels whose delay spreads and fluctuation rates are large. In the RLS and *i*DCD-RLS algorithms, $\lambda=1-2^{-9}$, $N_1=0$, $N_2=31$, and $N_3=16$; in the FDE-RLS algorithm, the block duration is set as 60 ms, $\lambda=1-2^{-9}$, $N_1'=0$, $N_2'=18$, and $N_3'=9$. The computationally efficient normalized least mean square (NLMS) algorithm, with a step size set at 0.5 through a trial-and-error process, is also incorporated for comparative evaluation. Additive white Gaussian noise is employed in the simulation.

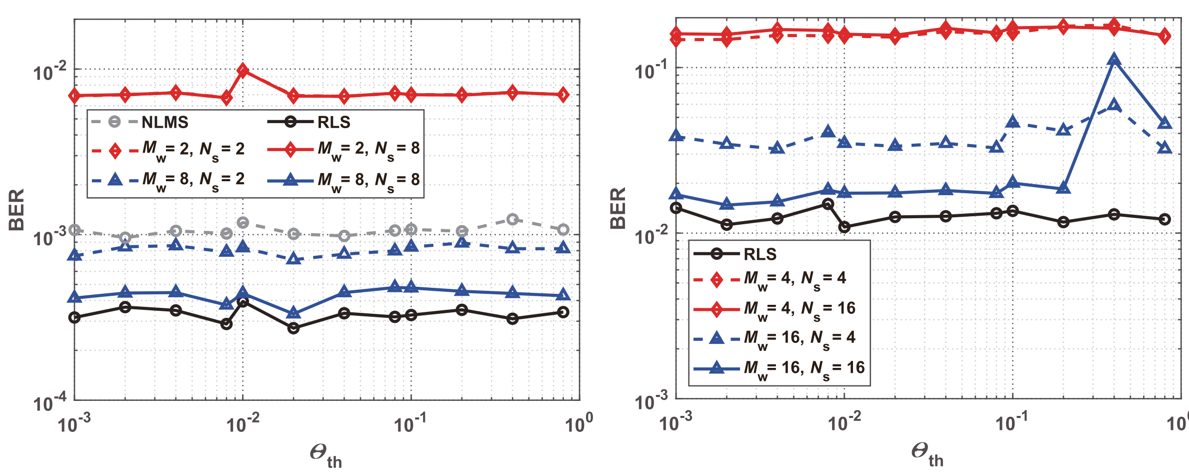
A. *Performance of iDCD-RLS-Based DFE*

We first investigate the threshold for phase correction in *i*DCD-RLS-based DFE. Monte Carlo simulations are carried out at the received SNR of 20 dB over 100 iterations for each threshold. In each iteration, a 2-s signal is modulated by QPSK. Different M_ws and N_ss are used in the simulations. The BERs and complexities for different thresholds are shown in Figs. 7 and 8, respectively, where N_CM denotes the numbers of CMs. As



(a) (b)

Fig. 6. Channel impulse responses generated by the simulator at transmission ranges of (a) 500 m and (b) 1000 m. The depths of the transmitter and the receiving element are 10 and 15 m, respectively. The starting and ending delays are marked by the white dashed lines, and the delay spreads are 9.9 and 15.7 ms.



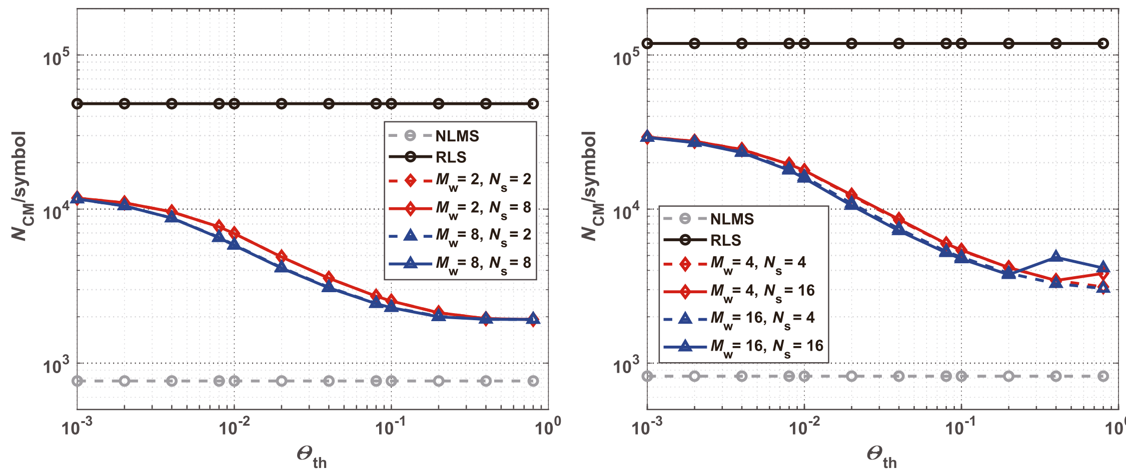
(a) (b)

Fig. 7. BER-$\Theta_{\rm th}$ performance curves in (a) *mild* and (b) *wild* channels with *i*DCD-RLS-based DFE.

expected, the algorithm performs better with lower complexity in *mild* channels than that in *wild* channels. On the whole, the bit number $M_{\rm w}$ contributes the most, and the larger value leads to better performance because of the higher resolution. The number of *successful* iterations $N_{\rm s}$ also helps improve the BER performance, as shown in Fig. 7. The larger $M_{\rm w}$ and $N_{\rm s}$ do not cost additional computational complexity, as shown in Fig. 8. Note that the signal is sampled at a rate of two samples per symbol (SPS) here.

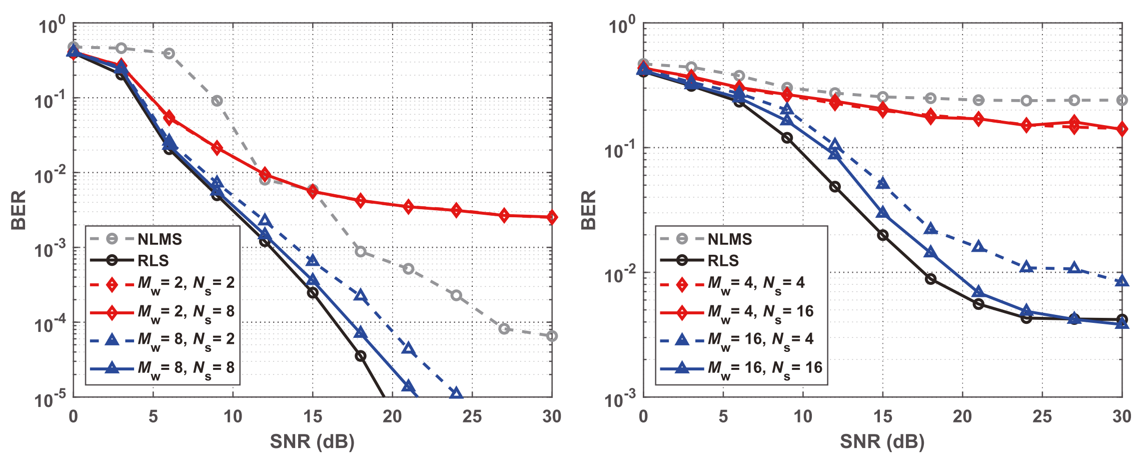
For the case of *mild* channels, the phase fluctuation is relatively small. The algorithm at, for example, $\Theta_{\rm th} = 0.8$, can even be implemented without a phase correction at $M_{\rm w} = 8$ and $N_{\rm s} = 8$. The corresponding BER is close to that of RLS-based DFE, as depicted in Fig. 7(a); meanwhile, the computational overhead of CMs is minimal, which is $11N_0$ and slightly higher than that of NLMS-based DFE, as shown in Fig. 8(a). For the case of *wild* channels, the phase fluctuation cannot be ignored. At $M_{\rm w} = 16$ and $N_{\rm s} = 16$, the performance deteriorates when $\Theta_{\rm th}$ exceeds 0.1 and is even worse than that of $N_{\rm s} = 4$, as shown in Fig. 7(b). Since the phase correction is not implemented for large accumulated phase increments, the correlation matrix does not update in the right way and results in larger phase estimate errors. The error rises further and requires a more frequent phase correction, leading to a heavier overhead of CMs, as shown in Fig. 8(b). Hence, an appropriate threshold would be $\Theta_{\rm th} = 0.1$. The corresponding number of CMs is around $17N_0$. The NLMS algorithm is not applicable in this scenario, and the corresponding BERs are not presented in Fig. 7(b).

Next, we evaluate the performance of the *i*DCD-RLS algorithm at different SNRs. The threshold is chosen as $\Theta_{\rm th} = 0.1$ according to the previous analysis. As shown in Fig. 9, the *i*DCD-RLS algorithm approximates the conventional RLS one at large $M_{\rm w}$s and $N_{\rm s}$s. In the *mild* channels, the performance



(a) (b)

Fig. 8. Complexity–$\Theta_{\rm th}$ curves in (a) *mild* and (b) *wild* channels with *i*DCD-RLS-based DFE.



(a) (b)

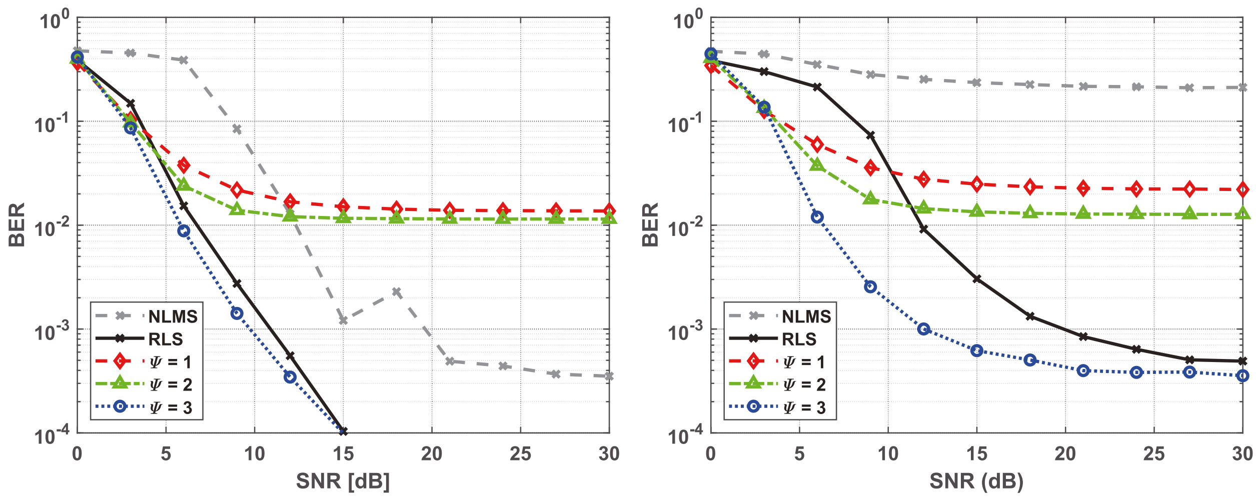
Fig. 9. BER–SNR performance curves in (a) *mild* and (b) *wild* channels with RLS- and *i*DCD-RLS-based DFEs.

gap is only 1.0 dB at the BER of 10^{-4} for $M_{\rm w} = 8$ and $N_{\rm s} = 8$. The performance of NLMS-based DFE is inferior to the *i*DCD-RLS-based one with a roughly 9-dB discrepancy, despite its slightly lower complexity. In the *wild* channels, $M_{\rm w}$ and $N_{\rm s}$ should be tuned up to 16 for a better performance. The gap is 2.0 dB at the BER of 10^{-2}. There exists an error floor due to the residual ISI when the SNR is higher than 20 dB.

B. Performance of FDE-RLS-Based DFE

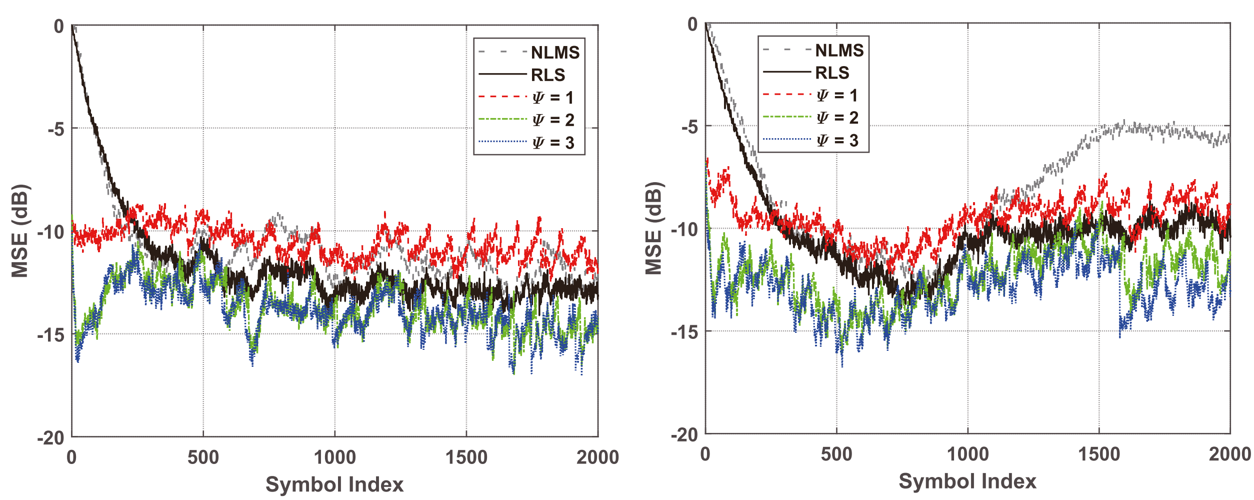
In the FDE-RLS-based DFE, we first examine the BER performance with different iterations. As shown in Fig. 10, the performance is improved with the increasing iterations. For $\Psi = 1$ and 2, the channel estimates and the symbol decisions are not reliable enough, and the performance approaches saturation when the SNR is higher than 10 dB in the two types of channels. Both the NLMS- and RLS-based DFEs outperform the FDE-RLS-based one. If $\Psi = 3$, the error floor is not observed in the *mild* channels, and the performance gap between the FDE-RLS and conventional RLS algorithms is 1.0 dB at the BER of 10^{-3}. Due to the residual errors from overlapping subblocks and mismatch between the blockwise and symbolwise equalizations, the FDE-RLS-based DFE was slightly inferior to the RLS-based one for SNRs higher than 15 dB. However, the performance gap can be further filled by more iterations at the sacrifice of higher complexity. In the *wild* channels, the FDE-RLS algorithm even outperforms the RLS algorithm in the investigated range of SNR. The error floor is lower at high SNRs for FDE-RLS-based DFE. Note that the complexity of the FDE-RLS algorithm changes little for different SNRs.

To better illustrate the performance of FDE-RLS-based DFE, we plot the MSEs with different iterations at the SNR of 15 dB, as shown in Fig. 11. For $\Psi = 1$ of both the *mild* and *wild* channels, MSE fluctuates with a period of 84 due to the mismatch among



(a) (b)

Fig. 10. BER–SNR performance curves in (a) *mild* and (b) *wild* channels with RLS- and FDE-RLS-based DFEs.

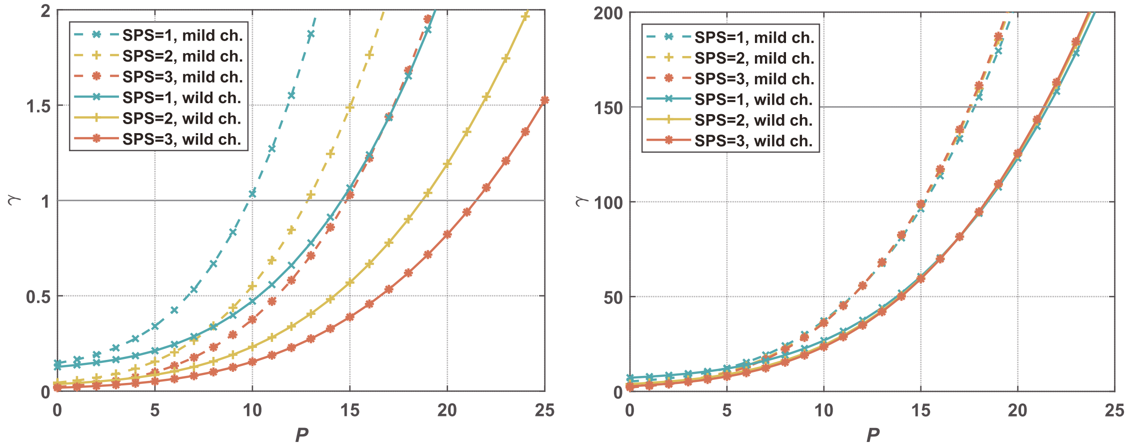


(a) (b)

Fig. 11. MSE performance of the RLS- and FDE-RLS-based DFEs. (a) In the *mild* channels, the average MSEs for data symbols in NLMS- and RLS-based DFEs are -11.4 and -12.5 dB, respectively, and the average MSEs in FDE-RLS-based DFE are -10.5, -13.3, and -13.5 dB for $\Psi = 1$–3, respectively. (b) In the *wild* channels, the average MSEs for data symbols in NLMS- and RLS-based DFEs are -7.4 and -10.6 dB, respectively, and the average MSEs in FDE-RLS-based DFE are -9.4, -12.1, and -12.8 dB for $\Psi = 1$–3, respectively.

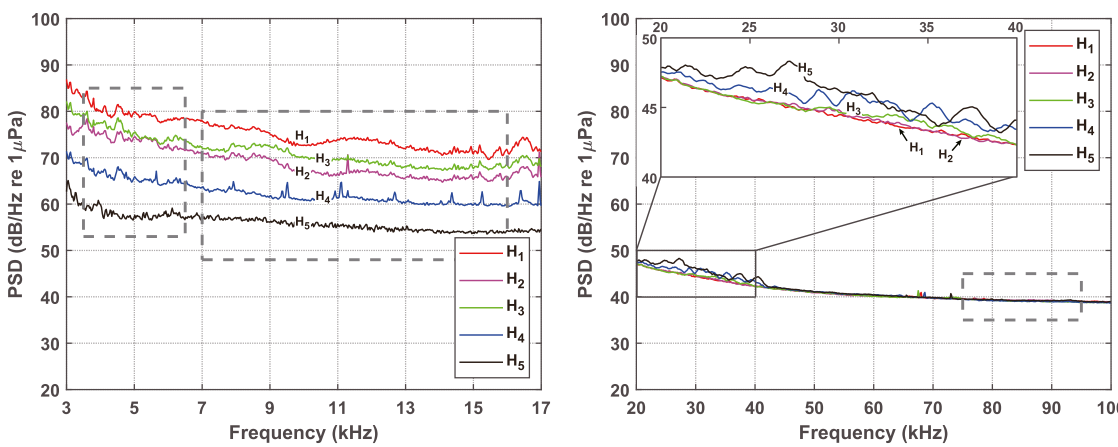
adjacent subblocks, whose lengths equal to the period. With the increase of iterations, the MSE is suppressed to a lower level, and the periodic phenomenon becomes less evident. For $\Psi = 2$ of the *wild* channels, the backshifting structure can be observed. At $\Psi = 3$, the MSE is smaller than the traditional RLS-based one, thus resulting in better performance. The MSE of the NLMS algorithm increases after 1000 symbols, resulting in decoding failure. In the *mild* channels, the performance of the FDE-RLS algorithm saturates at $\Psi = 2$.

To examine the complexity reduction of the FDE-RLS algorithm, we calculate the ratio between the CMs of FDE-RLS-based DFE and RLS-based (or NLMS-based) DFE for $\Psi = 3$. The ratio is designated as γ, and the smaller value of γ suggests a greater complexity saving. As shown in Fig. 12, the complexity of the FDE-RLS algorithm increases with the increasing number of dominant paths P in OMP. The saving is greater for the *wild* channels, which exhibit longer delay spreads, as shown in Fig. 12(a). In our simulations, P is set to 10, and the ratios are 55% and 23% in the two types of channels, respectively. When compared to NLMS-based DFE, the complexity of FDE-RLS-based DFE is 36 times in the *mild* channels and 24 times in the *wild* channels, as shown in Fig. 12(b). Improved OMP or other efficient algorithms can be adopted here for lower computational overhead in channel estimation, which is, however, beyond the scope of this article. On the other hand, a higher baseband sampling rate results in a greater complexity saving. It also



(a) (b)

Fig. 12. CM ratios (a) between FDE-RLS- and RLS-based DFEs and (b) between FDE-RLS- and NLMS-based DFEs. To achieve complexity reduction compared to RLS-based DFE ($\gamma < 1$), the maximum numbers of dominant paths P are 9, 12, and 14 for SPS = 1–3 in the *mild* channels, and 14, 18, and 21 in the *wild* channels, respectively. The resulting complexity is no more than 150 times that of NLMS-based DFE.



(a) (b)

Fig. 13. Noise PSDs in (a) SCS20 and (b) GoM17 experiments, where H_1–H_5 cover the water depth of 8–10.67 m in SCS20, and the water depth of 10–18 m in GoM17. The gray dashed boxes denote the bands of interest.

implies that the FDE-RLS algorithm would be preferred in UWA channels with long delay spreads.

VI. Experimental Results

In this section, we present the results from two experiments, the South China Sea Experiment in August 2020 (SCS20) and the Gulf of Mexico Experiment in August 2017 (GoM17). In each experiment, a receiver array with five receiving elements H_1–H_5 was deployed in the shallow layer. The power spectral densities (PSDs) of ambient noise within the band of interest are plotted in Fig. 13. In the SCS20 experiment, we transmitted lower frequency signals in two bands, which exhibited lower PSDs from the deeper receiving elements. In the GoM17 experiment, higher frequency signals were transmitted, and the effect of depth variation can be disregarded. In both the experiments, the noise PSD from each receiving element can be considered white within the band of interest. The channels in these two experiments fall into the categories of *mild* and *wild* channels, respectively. The *i*DCD-RLS- and FDE-RLS-based DFE algorithms are examined in terms of BERs and numbers of CMs.

A. Results in the SCS20 Experiment

The experiment was conducted off the Paracel Islands in August 2020. The water depth was around 1000 m. A 5-kHz transducer T_1 and a 11.5-kHz transducer T_2 were hung from the Research Vessel (R/V) *Fengwang* and deployed at the depth of 10 m. A receiver array with five receiving elements H_1–H_5 was hung from the *R/V Pengyuan* and covered the water depth of 8–10.67 m with an element spacing of 0.67 m. During the transmissions, *R/V Pengyuan* remained nominal stationary. *R/V Fengwang* initially traveled away from *R/V Pengyuan* and transmitted signals at five sites ranging from 3.8 to 8.0 km (denoted as

S1–S5) and then moved back and transmitted signals at five sites ranging from 6.8 to 3.2 km (denoted as S6–S10). The relative speed was around 3.6 m/s during the S1–S5 transmissions and 1.2 m/s during the S6–S10 transmissions.

In each packet, two 0.2-s linear frequency-modulated (LFM) signals were placed before and after the data blocks, each of which included a 511-symbol m-sequence and a 2048-symbol QPSK-modulated sequence. The LFM signals were used for bulk Doppler estimation and coarse estimation of channel length. The gaps before and after the LFM signals were long enough to avoid interference. The m-sequence in each block was used for symbol synchronization and training of the RLS algorithm. Ten-block signals at a symbol rate of 2.67 kilo-symbols per second (ksym/s) were transmitted over a 3-kHz bandwidth for the 5-kHz transducer, and 30-block signals at a symbol rate of 8 ksym/s were transmitted over a 9-kHz bandwidth for the 11.5-kHz one. Pulse shaping and matched filtering employed the root raised cosine filter with a roll-off factor of 0.125. The total duration of a packet was 10.9 s.

At the receiver, signals from the top four receiving elements were recorded at a sampling rate of 96 kHz and transformed to the baseband with SPS $= 2$. Resampling was then performed to suppress the bulk Doppler. Throughout all the RLS-based DFEs, the forgetting factor was set as $\lambda = 1–2^{-11}$, and the proportional and integral tracking constants in PLL were set as 0.001 and 0.0001, respectively. In the *i*DCD-RLS algorithm, the threshold of phase correction was set as $\Theta_{\text{th}} = 0.1$. Indeed, a higher threshold also worked for the equalizer as the phase variation was small. In the FDE-RLS algorithm, the number of iterations was set as $\Psi = 1$ or 2, and the signal was partitioned into 50-ms subblocks with an overlapping rate of $\alpha = 30\%$. Since the sparsity of channels was exploited in the FDE-RLS algorithm, sparse penalties, i.e., l_0 and l_1 norms, were integrated in the RLS and *i*DCD-RLS algorithms for a fair comparison. These sparse algorithms were referred to as RLS-l_0, RLS-l_1, *i*DCD-RLS-l_0, and *i*DCD-RLS-l_1. In the RLS-l_0 algorithm, the sparse function is approximated by [29], [30]

$$f(\mathbf{w}) = \|\mathbf{w}\|_0 \approx \sum_{i=0}^{N_0-1} \left(1 - e^{-\epsilon|w(i)|}\right) \tag{38}$$

where ϵ is a constant and the subscript of $\mathbf{w}_{n-1}$ is dropped for the sake of simplicity. By exploiting the first-order Taylor series expansion in the subgradient of (38) with respect to $\mathbf{w}$, one obtains the sparse updating term $\mathbf{\Phi}^{-1}\partial f(\mathbf{w})/\partial \mathbf{w}$ with

$$\frac{\partial f(\mathbf{w})}{\partial \mathbf{w}} \approx \begin{cases} \epsilon \mathrm{sgn}(w(i)) - \epsilon^2 w(i), & |w(i)| \leqslant 1/\epsilon \\ 0, & \text{otherwise.} \end{cases} \tag{39}$$

In the RLS-l_1 algorithm, $f(\mathbf{w}) = \|\mathbf{w}\|_1$, and $\partial f(\mathbf{w})/\partial \mathbf{w} = \mathrm{sgn}(\mathbf{w})$. The key parameters can be found in Table IV. The parameter selection was guided by three considerations.

1) Since the channels in the SCS20 experiment are categorized as *mild* channels, typical parameter values of K, λ (approximate to 0.9995), K_{f1}, and K_{f2} were used to track the slowly time-varying channels.

TABLE IV
PARAMETERS OF RLS-BASED DFES IN SCS20

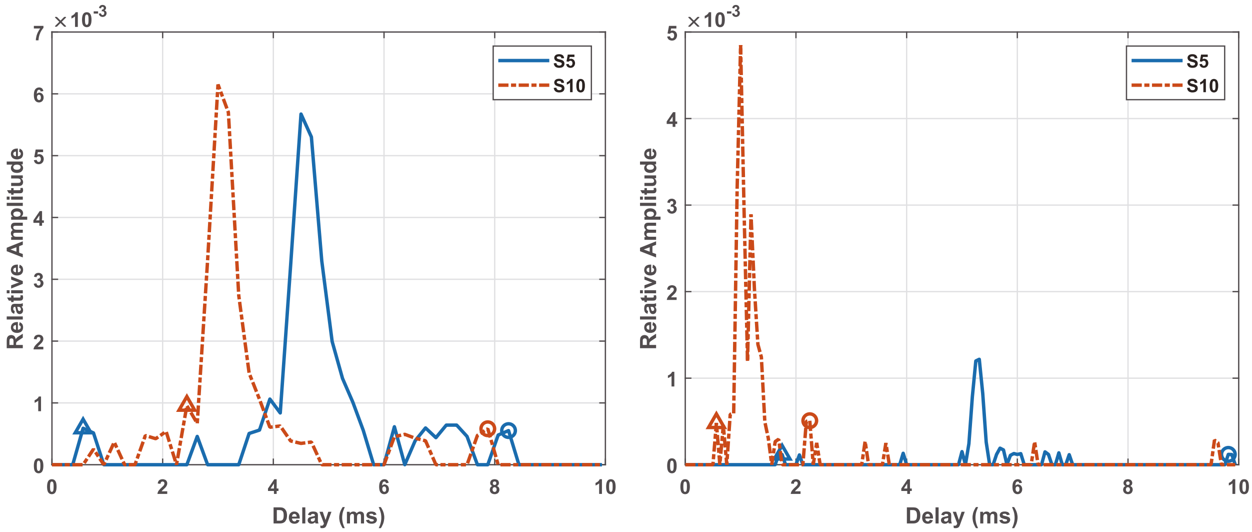
Parameters	Descriptions	Values
λ	Forgetting factor	$1 - 2^{-11}$
K_{f1}	Proportional tracking constant	0.001
K_{f2}	Integral tracking constant	0.0001
Θ_{th}	Threshold of phase correction	0.1
M_{w}	Number of bits for $\Delta\mathbf{w}_{n+1}$ in DCD	8
N_{s}	Number of *successful* iterations in DCD	8
Ψ	Number of iterations in FDE-RLS-based DFE	2
T_b	Block duration in FDE-RLS	0.05 s
α	Block overlapping rate in FDE-RLS	30%

2) The parameters related to the *i*DCD-RLS algorithm, including Θ_{th}, M_{w}, and N_{s}, were kept consistent with those used for *mild* channels in simulations.
3) For the FDE-RLS algorithm, the number of iterations (Ψ) was set as 2, and the block duration (T_b) was set as four to six times the maximum channel delay spread to achieve a good tradeoff between the accuracy of channel estimates and the complexity of channel equalization. The block overlapping rate was chosen such that $N_{\text{ov}}/2 \geqslant N_2' - 1$ to mitigate precursor interference.

The channel estimates are depicted in Fig. 14. The multipath starting and ending delays were estimated by the adaptive thresholding algorithm with the LFM signals. In comparison with the lower frequency channel, the higher frequency one exhibited lower amplitudes and more concentrated multipath components due to higher attenuation. During short-range transmissions, the amplitudes of the lower band channel were close to those of the higher band one, but the delay spreads were approximately three times as extensive, as observed in Table V(b). However, for long-range transmissions, the delay spreads were extended due to the larger sound ray difference, particularly in the case of the higher band channels. The considerably reduced amplitudes also indicate higher attenuation in higher frequency transmissions.

The results indicate that the lower frequency transmissions achieved better performance than higher frequency transmissions, as shown in Table V(a). The difference was attributed to the lower received SNRs in higher frequency transmissions, which was caused by higher attenuation. In addition to the received SNR, channel length also significantly affected performance. During long-range transmissions, such as S4 and S5, the low SNRs and long channel lengths resulted in high estimation errors of (2) and, therefore, high BERs. Moreover, most of the lower and higher frequency transmissions exhibited similar complexities, as shown in Table V(b). Although a higher symbol rate was adopted in the higher frequency transmissions, the channel lengths were comparable to those in the lower frequency transmissions as described in the previous analysis.

In terms of equalizers, the l_0-based equalizers outperformed the l_1-based counterparts, demonstrating lower BERs and reduced complexities in both the RLS- and *i*DCD-RLS-based DFEs. In comparison with the traditional RLS-based DFE, the *i*DCD-RLS-based DFE achieved similar performance but with



(a) (b)

Fig. 14. Typical channel estimates between the transmitter and H_2 in S5 and S10 transmissions in (a) 5- and (b) 11.5-kHz transmissions. The bulk Doppler shifts have been compensated for using LFM signals. The starting and ending delays are marked by triangles and circles, respectively.

TABLE V

BER PERFORMANCE AND COMPLEXITY FOR SIX RLS-BASED DFE SETS IN SCS20: (IA) RLS-l_0, (IB) RLS-l_1, (IIA) lDCD-RLS-l_0 WITH $M_\mathrm{w} = 8$ AND $N_\mathrm{s} = 8$, (IIB) lDCD-RLS-l_1 WITH $M_\mathrm{w} = 8$ AND $N_\mathrm{s} = 8$, (IIIA) FDE-RLS WITH $\Psi = 1$, AND (IIIB) FDE-RLS WITH $\Psi = 2$

(a)

Sites	5-kHz band							11.5-kHz band						
	SNR	IA	IB	IIA	IIB	IIIA	IIIB	SNR	IA	IB	IIA	IIB	IIIA	IIIB
S1	33.0 dB	3.54	3.54	3.56	3.42	2.10	0.27	32.0 dB	4.65	4.65	4.24	4.19	2.34	1.69
S2	29.5 dB	0.27	0.27	0.29	0.27	0.02	0	26.4 dB	4.00	4.00	3.74	3.68	2.82	1.81
S3	28.2 dB	0	0	0.02	0.02	0.05	0	26.1 dB	4.44	4.49	3.87	4.00	2.76	1.44
S4	24.4 dB	4.37	4.42	4.69	4.49	2.29	0.66	16.6 dB	31.75	34.65	36.29	53.91	36.74	20.34
S5	19.9 dB	11.18	11.72	11.50	11.28	12.13	6.20	8.6 dB	53.57	56.70	503.17	508.76	80.79	60.17
S6	25.8 dB	0	0	0	0	0	0	24.8 dB	3.06	3.07	2.83	2.67	2.12	1.54
S7	26.4 dB	0	0	0.02	0.02	0.22	0	15.1 dB	3.64	3.67	3.54	3.45	3.96	2.69
S8	27.0 dB	0.54	0.59	0.78	0.54	3.17	1.25	25.3 dB	3.23	3.25	2.97	3.04	1.90	1.60
S9	28.3 dB	0.05	0.05	0.05	0.05	0	0	25.9 dB	2.86	2.87	2.66	2.69	1.43	1.44
S10	24.5 dB	0.15	0.15	0.15	0.12	0.05	0	13.1 dB	9.72	9.90	10.33	10.18	9.70	5.75

(b)

Sites	5-kHz band							11.5-kHz band						
	Spreads	IA	IB	IIA	IIB	IIIA	IIIB	Spreads	IA	IB	IIA	IIB	IIIA	IIIB
S1	7.5 ms	42.41	69.80	5.93	6.05	12.25	24.35	2.2 ms	33.81	55.57	5.09	5.17	13.12	26.16
S2	5.6 ms	26.99	44.29	4.65	4.56	12.25	24.43	2.2 ms	34.26	56.32	5.19	5.44	13.14	26.18
S3	7.5 ms	46.54	76.65	6.02	5.91	12.25	24.35	2.2 ms	34.71	57.08	5.18	5.39	13.20	26.24
S4	10.1 ms	82.94	137.00	8.60	9.14	12.28	24.20	5.1 ms	177.61	294.29	13.40	13.33	15.01	29.27
S5	9.6 ms	70.71	116.75	7.82	8.34	12.17	24.15	7.9 ms	413.64	686.70	96.46	100.18	15.64	30.82
S6	10.3 ms	74.69	123.32	7.66	7.92	11.99	23.89	2.4 ms	41.90	68.97	5.63	6.11	13.20	26.30
S7	7.3 ms	41.90	68.97	5.82	5.85	12.22	24.32	1.6 ms	17.51	28.67	3.66	4.05	12.76	25.51
S8	8.8 ms	60.08	99.10	7.13	7.62	12.16	24.17	2.8 ms	50.86	83.81	6.34	6.66	13.43	26.70
S9	6.8 ms	34.26	56.32	5.13	5.22	12.23	24.37	2.3 ms	34.26	56.32	5.04	5.24	13.14	26.18
S10	5.8 ms	26.99	44.29	4.49	4.54	12.25	24.43	2.1 ms	34.25	56.32	5.26	5.72	13.08	26.07

(a) average BERs ($\times 10^{-3}$). (b) N_CM per symbol ($\times 10^3$).

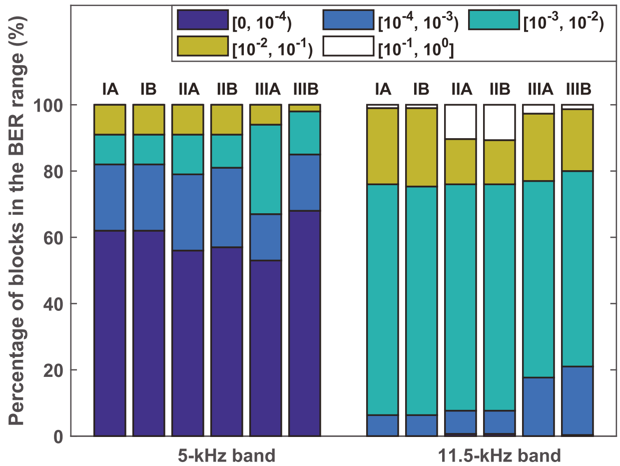


Fig. 15. Performance comparison among RLS-based DFEs in SCS20.

much lower complexity. For channels with large delay spreads, such as the lower frequency ones in S4–S6 transmissions, the complexity was reduced by at least tenfold. The FDE-RLS-based DFE exhibited superior performance while maintaining complexity levels between the other two DFEs. It is worth noting that it also worked efficiently for the higher frequency transmissions in S5, whose channels exhibited low SNRs and long delay spreads.

To better demonstrate the performance among different RLS-based DFEs, we compare the BER benchmark in terms of the percentage of all the packets that fall in the specified BER ranges, as shown in Fig. 15. For the lower frequency transmissions, the smaller attenuation led to larger received SNRs in comparison with the higher frequency transmissions as the phase correction worked in *i*DCD-RLS-based DFE. As a consequence, it achieved similar performance as the RLS-based one. The performance of higher frequency *i*DCD-RLS-based DFEs slightly degraded due to lower SNRs. However, for the FDE-RLS-based DFE, it outperformed the others when $\Psi = 2$ since the channel shortening has simplified the structure of channel and produced a less contaminated signal for the followed RLS algorithm. For higher frequency transmissions, one iteration also achieved comparable performance.

B. Results in the GoM17 Experiment

The GoM17 experiment was conducted off the Alabama cost for high-frequency transmissions in August 2017. The water depth was 20 m. An 85-kHz transducer T_0 was submerged at a depth of 12 m, hung from *R/V Wilson*. It was anchored at four different ranges, i.e., 200, 400, 600, and 1000 m (denoted as G1–G4), away from the moored receiver. The five-element receiver array H_1–H_5 covered the water depth of 10–18 m with an element spacing of 2.0 m. During the transmissions, both the transmitter and the receiver were stationary.

In each transmission, one to three packets were transmitted, each of which included a QPSK-modulated sequence with a symbol length of 84857. LFM signals and gaps were also inserted before and after the sequence. The signal was transmitted

TABLE VI
RECEIVED SNRs AND MSEs OF FDE-RLS-BASED DFE AT FOUR COMMUNICATION RANGES IN GoM17

Sites	G1	G2	G3	G4
Received SNRs (dB)	41.1	39.5	34.3	25.7
MSEs for $\Psi = 1$ (dB)	–7.9	–8.0	–8.3	–8.6
MSEs for $\Psi = 2$ (dB)	–11.6	–10.7	–10.9	–10.9
MSEs for $\Psi = 3$ (dB)	–12.5	–11.4	–11.5	–11.3

over a 20-kHz bandwidth at a symbol rate of 16 kHz. Pulse shaping and matched filtering employed the root raised cosine filter with a roll-off factor of 0.25. The total duration of a packet was 6.0 s.

At the receiver, the processing procedures were similar to those in SCS20. The sampling frequency was 512 kHz, and the baseband sampling rate was SPS $= 2$. Throughout all the RLS-based DFEs, the forgetting factor was set as $1–2^{-12}$, and the tracking constants in PLL were set as 10^{-4} and 10^{-5}. In RLS- and *i*DCD-RLS-based DFEs, demodulation results indicated that the MSE did not converge to a low level due to the long filter lengths that exceeded 1600 (see Fig. 16 and the corresponding analysis). All the packets failed in demodulation when using these two equalizers. However, the FDE-RLS-based DFE still worked for the packets with received SNRs higher than 25 dB. Here, we focus on the FDE-RLS-based DFE and present the corresponding results. The maximum number of iterations was set as 3, and the signal was partitioned into 45-ms subblocks with an overlapping rate of 30%. In each subblock, the impulse responses were estimated by compressive sampling matching pursuit [31].

The estimated channel impulse responses for G1 and G4 transmissions are presented in Fig. 16. The delay spreads of these channels were all around 10 ms, although the multipath structures exhibited different features. When taking into account the symbol rate of 16 ksym/s with SPS $= 2$ and five receiving elements, the length of each channel estimate was more than 1600. In short-range transmissions, despite heavy scattered micropaths, most of the reflection clusters were distinguishable from the others, as depicted in Fig. 16(a). In long-range transmissions, the smaller grazing angles resulted in the micropaths spanning in a wider range of spread, as shown in Fig. 16(b). As a consequence, the direct path and the specular-reflected multipaths dominated the arrivals. The corresponding coherence time then became larger with the increasing transmission range, with the highest observed in G4 transmission, as depicted in Fig. 16(c). Here, the channel correlation coefficients ρ were calculated according to [25].

The communication results of the FDE-RLS-based DFE are shown in Fig. 17, where the output SNRs and uncoded BERs are presented. When $\Psi = 1$, the receiver did not work as the filter coefficients of channel shortening in each subblock were calculated from the outdated impulse responses in previous subblock. The MSEs were consistently above -10 dB, as detailed in Table VI. With the increasing number of iterations, the demodulation performance was improved as the overlapping

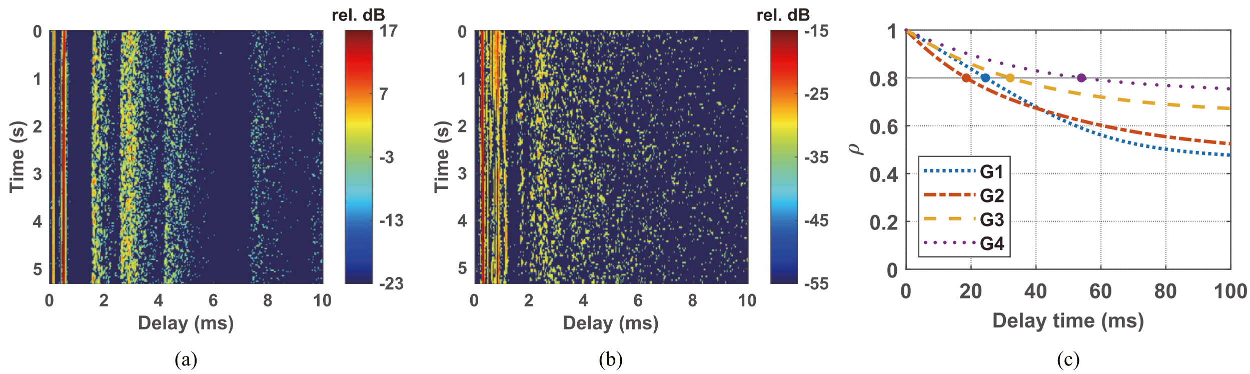


Fig. 16. Typical channel estimates between T_0 and H_3 in (a) G1 and (b) G4 transmissions. (c) Channel correlation coefficients in GoM17. The bulk Doppler shifts have been compensated for using LFM signals. The coherence times of impulse responses were 24.5, 18.6, 32.0, and 54.6 ms for the four transmission ranges, respectively.

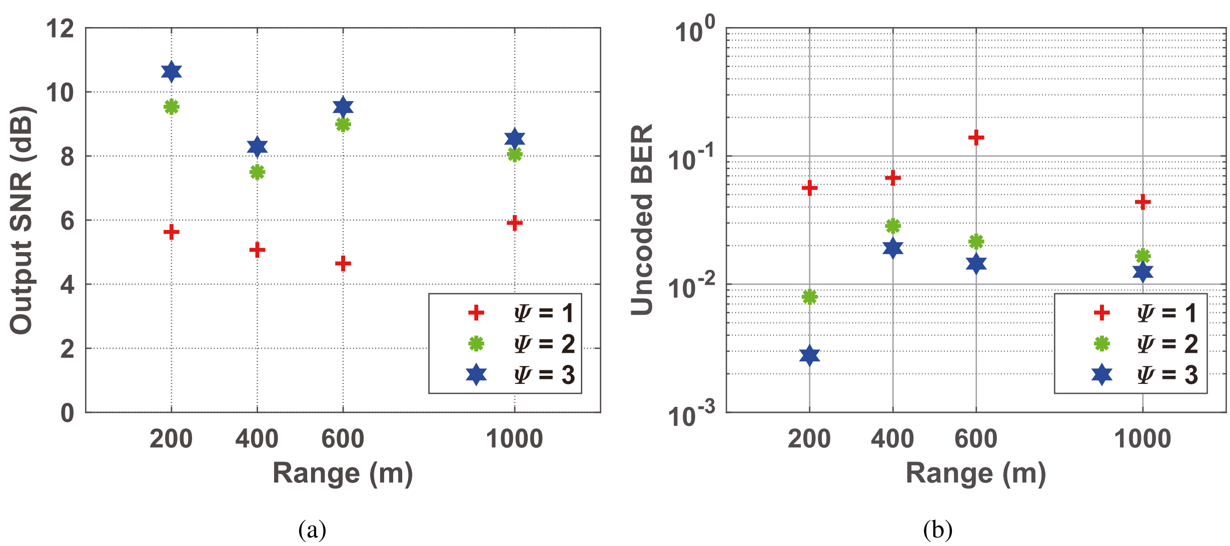


Fig. 17. Demodulation results of the FDE-RLS-based receiver with different iterations in GoM17. (a) Output SNRs. (b) Uncoded BERs.

partitioning and weighting processing addressed the mismatch between the blockwise and symbolwise equalizations. The average output SNRs were improved by 3.2 and 3.9 dB for $\Psi = 2$ and $\Psi = 3$, respectively. The corresponding average uncoded BER decreased by 70% and 80%, and the average MSE decreased by 2.8 and 3.5 dB, as indicated in Table VI. Performance improvements beyond four iterations can be negligible. It is noteworthy that the minimum received SNR did not produce the worst communication performance, which is often the case in UWA communications. Instead, the worst performance was observed during G2 transmissions, where the channel fluctuation rate was the fastest, as shown in Fig. 16(c).

VII. Conclusion

To achieve an efficient equalization in UWA channels with long channel taps, we propose two strategies to reduce the complexity of RLS-based DFEs. The first one adapts the DCD algorithm to the UWA environment with the nonshift structure of input sequence. Partial updating coupled with thresholding is utilized to avoid full-scale and frequent phase compensation of the cross-correlation matrix. The second strategy exploits the iterative FD-DFE for channel shortening. The overlapping partitioning and weighting processing makes the subsequent symbolwise RLS algorithm consistent across the subblocks and iterations. Those two strategies have been validated via simulation and experimental data recorded from two experiments: SCS20 and GoM17.

Simulation results demonstrate that both the strategies are applicable to slow or fast time-varying channels with low complexity. The threshold of partial updating, $\Theta_{\rm th} = 0.1$, also works for SCS20. In that experiment, the performance of *i*DCD-RLS- and FDE-RLS-based DFEs was similar to that of traditional RLS-based one, whereas the complexity savings were 90% and 70% ($\Psi = 1$), respectively. In the GoM17 experiment, only the FDE-RLS-based DFE worked for the high-frequency UWA channels with long taps and fast variability, and $\Psi = 2$ provided a good tradeoff between the performance and complexity.

Appendix
Derivation of (20)

Substituting (12b) into (14) (with n being $n-1$) yields

$$\Delta \mathbf{c}_{n-1} = (\lambda - 1)\mathbf{c}_{n-2} + x^*(n-1)\mathbf{u}_{n-1}. \quad \text{(A1)}$$

Substituting (12a) into (15) (with n being $n-1$) yields

$$\Delta \mathbf{\Phi}_{n-1} = (\lambda - 1)\mathbf{\Phi}_{n-2} + \mathbf{u}_{n-1}\mathbf{u}_{n-1}^H. \quad \text{(A2)}$$

Then, substituting (A1) and (A2) into (18b) yields

$$\begin{aligned}\mathbf{a}_{n-1|n-2} &= \mathbf{a}_{n-2|n-2} + \Delta \mathbf{c}_{n-1} - \Delta \mathbf{\Phi}_{n-1}\hat{\mathbf{w}}_{n-2} \\ &= \mathbf{a}_{n-2|n-2} + \left[(\lambda - 1)\mathbf{c}_{n-2} + x^*(n-1)\mathbf{u}_{n-1}\right] \\ &\quad - \left[(\lambda - 1)\mathbf{\Phi}_{n-2} + \mathbf{u}_{n-1}\mathbf{u}_{n-1}^H\right]\hat{\mathbf{w}}_{n-2}. \quad \text{(A3)}\end{aligned}$$

The term $\mathbf{\Phi}_{n-2}\hat{\mathbf{w}}_{n-2}$ can be replaced with (13) (with n and n' both being $n-2$), and (A3) can be written as

$$\begin{aligned}\mathbf{a}_{n-1|n-2} &= \mathbf{a}_{n-2|n-2} + \left[(\lambda - 1)\mathbf{c}_{n-2} + x^*(n-1)\mathbf{u}_{n-1}\right] \\ &\quad - (\lambda - 1)\left(\mathbf{c}_{n-2} - \mathbf{a}_{n-2|n-2}\right) - \left(\hat{\mathbf{w}}_{n-2}^H\mathbf{u}_{n-1}\right)^*\mathbf{u}_{n-1} \\ &= \lambda \mathbf{a}_{n-2|n-2} + \left[x(n-1) - \hat{\mathbf{w}}_{n-2}^H\mathbf{u}_{n-1}\right]^*\mathbf{u}_{n-1}. \quad \text{(A4)}\end{aligned}$$

Then, (20) can be obtained by replacing n with $n+1$ in (A4).

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